bb-amp — Design Document

ai-ee v1 pipeline

generated 2026-08-17 23:02:48

Contents

1 Board and Run Metadata

Field	Value
board	bb-amp
workspace	`boards/bb-amp`
phase	P9
gate status	all 6 recorded gates pass
state created	2026-08-16T17:25:38
state updated	2026-08-17T23:02:25

2 Overview

bb-amp brief (owner's words, verbatim)

learning block-basics: a sensor amplifier. Input a differential 0-20 mV signal on a screw terminal, from an off-board bridge sensor with its own excitation. Output 0-3.3 V single-ended on a screw terminal, DC to 1 kHz. Powered from an external 3.3 V rail on its own screw terminal. Workspace boards/bb-amp.

Mode (resolved at P0 by state.py mode)

- token: `learning block-basics:`
- target: `block-basics`
- scope tier: `block-only`
- binding level: `canonical` (geometry is an OUTPUT of the placement)

- stage under study: none

3 Requirements

bb-amp - requirements

Source: `brief/brief.md` (owner's verbatim brief). Written at P0 by the requirements-analyst. Nothing here selects a part, a topology or a value: the unknowns that would decide those are section 9, and they are for the owner.

1. Function

bb-amp is a sensor-amplifier signal chain on a bare board. It takes a differential 0-20 mV signal from an OFF-BOARD bridge sensor (the sensor carries its own excitation, so this board provides NO excitation circuitry), amplifies it, and presents 0-3.3 V single-ended, referenced to board ground, over DC to 1 kHz. It is powered from an external 3.3 V rail. The nominal transfer is 165 V/V - that is 3.3 V / 20 mV restated, not a topology decision; the exact gain follows from the answers to Q5, Q6 and Q10 and belongs to P2/P4.

Build mode (resolved at P0 by `state.py mode`; do not re-derive):

```
| dial | value |
|---|---|
| token | ‘learning block-basics:‘ |
| mode / target | ‘block-basics‘ |
| scope tier | ‘block-only‘ |
| binding level | ‘canonical‘ - geometry is an OUTPUT of the placement |
| stage under study | none |
```

Scope consequences every later phase and both reviewers should read from here:

- ON the board: the amplifier block's active part(s), exactly the support components their datasheets require for correct operation at the stated operating point (decoupling, the reference the amplifier's own single-supply operation needs, gain-setting components), plus one input interface, one output interface and the power entry - the three screw terminals the brief names.
- NOT on the board, and never a reviewer finding at `block-only`: protection (TVS/ESD, reverse polarity, fusing, OVP/OCP), filtering the datasheet does not require, indicators, test-points beyond the block's own measurement need, config straps or DNP options, any second rail or second IC the block does not need, mechanical and enclosure features beyond mounting the bare board.
- NOT relaxed by the mode: every electrical spec, every gate, the P2/P3 coverage checks and the research they trigger, DFM, the datasheets' own requirements, and section 8. A mode relaxes geometry, cost and packaging only.

2. Interfaces

> **POLE ORDER CORRECTED AT P6.** J1's physical poles are **1 = IN-, 2 = IN+, 3 = GND**. J1's poles and the AD8226's input pin order were reversed, forcing the input pair to cross; swapping poles 1 and 2 uncrosses it and lets the pair route fully planar and mirror-symmetric with zero vias. Nothing electrical changed - IN+ still drives U1 pin 4 and the polarity convention is unchanged; only which physical pole carries IN+ moved. Any text below that reads `IN+ / IN- / GND` predates that swap. The board silk and the netlist are authoritative.

Three screw terminals, all named by the brief. They are the block's one input, one output and its power entry - in scope at `block-only`.

ref	interface	signal	electrical
J1	input	differential from the off-board bridge sensor	0-20 mV nominal full scale, DC to 1 kHz; common-mode level UNKNOWN -> Q1; pole count -> Q2
J2	output	single-ended, referenced to board GND	0-3.3 V nominal full scale, DC to 1 kHz; achievable span -> Q6; load -> Q8
J3	power	external 3.3 V rail + GND	see section 3

- **J1**, recommended 3 poles (IN+, IN-, GND) pending Q2. Two poles only is a floating input: an amplifier needs a DC return for its input bias currents and a defined common-mode reference, and without them the output is not predictable. This is the single riskiest unknown on the board.
- **J2**, 2 poles (OUT, GND). The output terminal is also the block's measurement point, so no separate test points are added.
- **J3**, 2 poles (+3V3, GND). One ground node on the board; J1/J2/J3 grounds are the same net.
- No connector standard is imposed by the brief. ASSUMED: pluggable or fixed screw terminal blocks on a 3.5 mm or 5.08 mm pitch accepting roughly 24-16 AWG wire, top side, silkscreened IN+ / IN- / GND / OUT / +3V3. The exact part is P3's call.
- ASSUMED: IN+ above IN- gives a positive output (polarity convention).
- ASSUMED: no ESD/TVS, no series protection and no anti-alias or band-limiting filter beyond the amplifier's own response - protection and filtering are excluded at `block-only`. Response shape at the top end is Q9.
- ASSUMED: sensor cable 1 m or shorter, shielded twisted pair, shield landed on J1's GND pole if Q2's default holds. No cable is supplied with the board.
- The board does NOT excite the bridge and has no excitation output: the brief states the sensor brings its own.

3. Power

- Single external rail, 3.3 V, entering on J3. No on-board regulation, no second rail, no battery, no charging, no power switch (a second rail is excluded at `block-only`).
- Rail tolerance and the current this board may draw: UNKNOWN -> Q10. ASSUMED for planning: 3.3 V +-5%.
- **GUESS (not a spec):** total board current under 10 mA. A precision single-supply amplifier plus whatever sets its reference is typically a few mA; P2/P3 replace this guess with the real datasheet numbers.
- In scope as datasheet-required support, not as "filtering": local decoupling at every amplifier supply pin as its datasheet specifies, and the reference node a single-supply amplifier needs in order to define its output zero.
- Consequence that drives the design: with only 0 V and 3.3 V available, the rail bounds BOTH the input common-mode range (Q1) and the output swing (Q6). Those two questions are the single-supply headroom problem, and they are not relaxable - they are electrical specs.

4. Environment

- Bench use, indoors, 0-50 C operating (block-only default; taken, not asked).
- No enclosure. No ingress rating. No vibration, shock or drop requirement. Non-condensing humidity; no conformal coating.
- Storage/handling: ordinary lab conditions. No altitude or pollution-degree requirement follows from a 3.3 V board.
- Accuracy consequence: drift across 0-50 C, not initial error, is what the accuracy target in Q7 mostly has to buy.

5. Size and mounting

No HARD cap. No dimension is stated in the brief, and none is imposed here. Under binding `canonical` the board size, aspect and outline are OUTPUTS of the placement, so any number that appeared here would be a PREFERENCE only - RELAXABLE (canonical) - and there is nothing to mark, because nothing is stated.

The canonical flow is mandatory and its order is load-bearing:

1. P5 `board_init --outline auto` - generous provisional room. `board_init` REFUSES a fixed outline under this binding; guessing a final size here binds placement to a number nobody has earned. 2. P6 place to the canonical layout for this block, gate `place`. 3. `board_edit --outline fit --margin M` - the board becomes what the placement needs. Re-run `planes_gen` if it GREW. 4. P7 route.

- Mounting: none required for bench use. Mechanical is excluded at `block-only` (mounting the bare board is all that is in scope), so the absence of mounting holes is not a reviewer finding; if the earned outline leaves corner room, plain M3 clearance holes may be added at P5/P6.
- Height: no limit - no enclosure. Terminal blocks are the tallest parts.
- Layer count is P2's decision under the "fewest honest layers" default; a ground reference under a 165 V/V DC signal chain is the reason to expect two.

6. Quantity and budget

- Build quantity 5 (block-only default; taken, not asked).
- No target unit cost is stated. Cost minimal, within the accuracy answered in Q7: a 12-bit-usable chain and a 16-bit chain are different part classes and different money, which is why Q7 is asked rather than assumed cheap.
- Standard JLC process only; no NRE beyond it; no exotic stackup, no controlled impedance, no gold fingers.

7. Assembly

- JLC PCBA, single-sided (top) assembly (block-only default; taken, not asked).
- ASSUMED: the SMT parts are placed by JLC; the three screw terminals are through-hole in most catalogs and JLC's standard SMT service does not place them - default is that the owner hand-solders three terminal blocks on the top side, unless P3 finds suitable SMD terminal blocks. Either resolution is acceptable; P3 records which.
- Lead-free process, fab-default finish and soldermask. No conformal coating, no potting, no cable assemblies, no enclosure.

8. Compliance and safety flags

Recorded plainly: **none of the flagged hazard classes apply to this board.**

```
| flag | applies?  | why |
|---|---|---|
| mains voltage | no | powered from an external 3.3 V bench rail (SELV); nothing
on the board is mains-referenced |
| battery / charging | no | no battery, no cell holder, no charge circuit |
| motors / inductive loads | no | output drives a measurement input, not a load |
| voltage above 30 V | no | 3.3 V is the highest potential anywhere on the board
|
| current above 3 A | no | expected board current is milliamps (see section 3) |
| RF transmit | no | no radio, no intentional radiator |
```

Two conditions would re-open this section, and both hang off the off-board sensor that this board cannot see:

- If the bridge excitation named in Q1 is above 30 V, J1 becomes a hazardous-voltage interface and section 8 must be re-answered before P2.
- If the sensor is mounted on equipment whose chassis sits at mains potential, the input cable carries that potential onto this board; that is an isolation requirement, not an amplifier requirement, and it is out of this board's current scope.

No mode grants silence here: if either condition is true, say so in the answers and the pipeline stops to re-scope rather than guessing.

9. Open questions

Batched for one round of answers. Each is closed-form with a recommended default; taking every default is a valid answer.

1. **What voltage excites the bridge, and does that excitation share this board's ground?** Recommended default: 3.3 V excitation, sharing this board's ground, so both input wires sit near 1.65 V - comfortably inside what a 3.3 V-powered amplifier can accept. Why it matters: with 5 V or 10 V excitation (both common on load cells) the input wires sit near 2.5 V or 5 V; 5 V is OUTSIDE the input range of anything powered from 3.3 V and forces a different front end. Above 30 V it also changes section 8.

2. **Does the sensor cable bring a ground wire to this board - i.e. should the input terminal have 3 poles (IN+, IN-, GND) rather than 2?** Recommended default: yes, 3 poles, with the cable shield landing on that GND pole. Why it matters: with only two wires the input has no DC path for the amplifier's input bias currents and no defined common-mode level, and the output drifts to a rail. Two poles is only safe if the sensor's excitation return is already tied to this board's ground somewhere else - if so, say where.

3. **What is the bridge's nominal resistance (the impedance seen looking back into each input)?** Recommended default: 350 ohm, the common load-cell value. Anything from 120 ohm to about 10 k is unremarkable for a high-impedance front end; above 10 k the amplifier's own input current starts to add error and narrows the part choice.

4. **Must the output track the excitation (ratiometric), or is a fixed amplifier enough (absolute)?** Recommended default: absolute - a fixed gain, with accuracy resting on the excitation being stable. Why it matters: a bridge's output is proportional to its excitation, so a 1% excitation drift is a 1% reading error in the absolute case. Ratiometric operation removes that, but needs a sense wire from the off-board excitation supply, which is a second input interface the `block-only` scope does not carry - answering "ratiometric" is a scope change, not a tweak.

5. **Beyond the nominal 0-20 mV, what can actually appear at the input?** Recommended default: down to about -1 mV (a real bridge rarely reads exactly zero at rest) and up to

about +25 mV under overload, with the amplifier expected to clip gracefully and recover - no added protection parts, which are excluded at this scope. Why it matters: if slightly negative inputs are possible, the output must sit at a small positive pedestal (see Q6) or a negative zero reads as a hard 0 V and cannot be calibrated out.

6. **A board on a single 3.3 V rail cannot output exactly 0.000 V or exactly 3.300 V - a real amplifier stops short at each end. Is a guaranteed usable output of roughly 0.05 V to 3.25 V acceptable, with full scale defined inside it?** Recommended default: yes, accept the reduced span, and place the 0 mV point at a small positive pedestal (about 0.1 V) so Q5's negative readings stay visible. Why it matters: demanding the true rails requires a supply above 3.3 V or a negative rail - a second rail, which this scope excludes - so this is the one place where the brief's ”0-3.3 V” and its ”3.3 V rail” genuinely conflict, and the owner picks which one bends.

7. **How finely does the reading need to be resolved, and is the system calibrated?** Recommended default: usable to 12 bits over the output span (about 0.8 mV at the output, about 5 microvolts referred to the sensor), with zero and span calibrated by whatever reads the output, so initial offset and gain error are trimmed and only noise and 0-50 C drift remain. Why it matters: 8 bits makes almost any amplifier acceptable and cheap; 16 bits changes the part class, the error budget and the cost.

8. **What will the 0-3.3 V output drive, and over how much cable?** Recommended default: a high-impedance input (bench meter or ADC input, above 100 k) on a lead under 1 m and under about 1 nF of capacitance. Why it matters: a capacitive cable or a load that draws real current changes the output stage and can make an otherwise fine amplifier ring.

9. **”DC to 1 kHz” - should the response still be flat (within about 1%) at 1 kHz, or is 1 kHz where it has already fallen by about 30% (-3 dB)?** Recommended default: flat within about 1% at 1 kHz, which means designing the amplifier's bandwidth several times higher. Why it matters: it sets how fast the amplifier has to be at a gain of 165, which is a part-selection constraint.

10. **How accurate is the external 3.3 V rail, and how much current may this board draw from it?** Recommended default: 3.3 V +-5% and a 10 mA budget. Why it matters: a rail that sags to 3.0 V shrinks the usable output span in Q6, and a tight current budget rules out some precision parts.

9a. ANSWERS (owner, 2026-08-16) - these are now REQUIREMENTS

Every question above is answered. Each is also a `state.py decision` at P0. Downstream phases design to THIS section, not to section 9's defaults.

Q	answer
1	Excitation **3.3 V, sharing this board's ground**. Input common-mode sits at **~1.65 V**.
2	**3-pole input terminal**. Physical pole order is **1 = IN-, 2 = IN+, 3 = GND** (corrected at P6 to uncross the input pair; see the note in section 2). Cable shield lands on the GND pole.
3	Bridge **350 ohm** nominal.
4	**Absolute** gain (NOT ratiometric). No excitation-sense input.
5	Input range to design for: **-1 mV to +25 mV**; beyond that the amplifier clips gracefully and recovers. No added protection parts.
6	**Reduced span with a pedestal.** Guaranteed usable output ~**0.05 V to 3.25 V**; 0 mV input maps to a small positive pedestal of about **0.1 V**. The brief's literal 0.000-3.300 V LOSES to the single 3.3 V rail.
7	**12-bit usable** over the output span: ~0.8 mV at the output, ~**5 uV referred to the input**. Zero and span calibrated downstream, so 0-50 C drift and noise are the real error budget.
8	Output drives **>100 kohm**, lead **<1 m**, **<1 nF**.
9	Response **flat within ~1% at 1 kHz** (1 kHz is inside the flat band, not the corner).
10	Rail **3.3 V +-5%**, board current budget **10 mA**.

What those answers make binding

- **Transfer function**: `Vout = Vped + G * Vin_diff`, with `Vped ~= 0.1 V`, and `Vin_diff = 20 mV` (full scale) landing INSIDE the guaranteed usable output top (~3.25 V). The gain follows from those two endpoints and is P2's to fix; it is close to but NOT equal to the brief's nominal 165 V/V, because the pedestal and the swing limit both eat span.

- **Pedestal source**: the ~0.1 V reference is a real circuit node. Whatever drives an instrumentation amplifier's REF input must be LOW impedance (or the amplifier must have a high-impedance REF by construction) - a bare resistor divider on a classic 3-op-amp in-amp REF pin degrades CMRR directly, and CMRR is the whole point of this block. P2 must say which mechanism it is using.

- **Bandwidth**: for a single-pole rolloff, 1% droop at 1 kHz puts the -3 dB corner near **7 kHz**. At a gain around 150-165 that is roughly **1 MHz of gain-bandwidth** in the gain path. Verify the rolloff-shape assumption; this is the dominant part-selection constraint and it disqualifies the lowest-power zero-drift in-amps.

- **Error budget** (12-bit usable, calibrated, 0-50 C): offset and gain error are trimmed out, so what must fit in ~5 uV RTI is **offset drift x 50 C**, gain drift, CMRR error against the 1.65 V common mode, and integrated noise from DC to the amplifier's own bandwidth. Noise density and 1/f corner matter here; initial offset does not.

- **Common mode**: 1.65 V, mid-rail, on a 3.3 V supply - benign, and the reason a single-supply in-amp front end stays viable. The input range is unipolar (0-20 mV nominal, -1 mV worst case), so the pedestal, not a mid-rail reference, sets the output zero.

Delegated authority The owner delegated the remaining engineering judgment for this run: decide in-run, optimizing for the LEARNING value of a basic sensor front end. H1-H4 are presented as records rather than blocking holds. The run stops at P9 per the `block-only` default - ordering stays a separate owner decision, so no irreversible or money-spending step is reached.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Build mode: mode block-basics; scope block-only; binding canonical	token 'learning block-basics:' in the brief; contract in reference/build-modes.md
P0	Q1 excitation: 3.3 V, sharing this board's ground -> input common-mode ~1.65 V	owner answer; keeps the inputs inside what a 3.3 V-powered in-amp accepts, so the canonical single-IC front end stays available (5 V or 10 V excitation would force a difference-amp with input attenuation)
P0	Q6 output span: usable ~0.05-3.25 V with a small positive pedestal (~0.1 V) at 0 mV input; the brief's literal 0.000-3.300 V loses	owner answer; a single 3.3 V rail cannot reach either rail exactly, and the alternative (second rail / charge pump) is excluded at block-only. The pedestal also keeps slightly-negative bridge readings (Q5, -1 mV) visible and calibratable. This is an ELECTRICAL spec resolution, not a mode relaxation

Phase	What	Why
P0	Q7 accuracy: 12-bit usable over the output span (~0.8 mV out, ~5 uV referred to input), zero and span calibrated downstream	owner answer; makes 0-50 C drift and noise the real error budget rather than initial offset, and keeps the part in the mainstream precision in-amp class rather than the 16-bit zero-drift class
P0	Q9 bandwidth: response flat within ~1% at 1 kHz (not -3 dB at 1 kHz)	owner answer; for a single-pole rolloff 1% droop at 1 kHz implies a -3 dB corner near 7 kHz, so at the full-scale gain the amplifier needs roughly 1 MHz gain-bandwidth - this is now the dominant part-selection constraint
P0	Q2/Q3/Q4/Q5/Q8/Q10 take their documented defaults: 3-pole input (IN+/IN-/GND, shield to GND), 350 ohm bridge, absolute (not ratiometric) gain, input -1 mV to +25 mV clipping gracefully, output into >100 k / <1 m / <1 nF, rail 3.3 V +-5% with a 10 mA budget	each default is the low-risk answer and none changes the topology; ratiometric in particular would add an excitation-sense input, i.e. a second input interface, which block-only scope excludes
P0	Owner delegated the remaining engineering judgment: decide in-run, optimizing for the best LEARNING value of a basic sensor front end. H1-H4 are presented as records and the run proceeds without blocking; the run stops at P9 (block-only default) so no irreversible or money-spending step is reached	owner instruction 2026-08-16 ('Make decisions yourself based on what will give the best learning for the basic system') plus three boards running in parallel; nothing before P10 is irreversible, and ordering stays a separate owner decision
P1	P1 roster: research-component-scout (block 'inamp') + research-reference-design (block 'bridge-front-end') only	single functional block. research-power-architect skipped: trivially powered (one external 3.3 V rail, no conversion, no second rail at block-only). research-interface-spec skipped: screw terminals are not standards-bound, so there is no spec to extract
P1	Reconciliation handed to P2: the scout's 'AD8226 REF grounded, pedestal injected at the second stage' does NOT survive the part's guaranteed output floor of (V-)+0.1 V - at G1=100 every input below +2 mV is in first-stage saturation, which is the bottom 10 percent of full scale. A positive, low-impedance-driven REF is unavoidable for any classic in-amp here	the pedestal exists to keep the -1 mV worst-case input visible (requirements 9a Q5/Q6); a stage-2 pedestal cannot recover information stage 1 has already clipped. Once a REF buffer is required anyway, single-stage at G~147 uses the same two ICs with fewer passives than the split

Phase	What	Why
P2	Topology: two-stage. U1 AD8226 at G1=39.9 (RG 1.27k, REF on a buffered 0.252 V node) -> U2B OPA2333-class RRIO at G2=3.49 whose gain resistor RETURNS TO /VREF, so Vout = Vref + G1*G2*Vdiff. U2A buffers the R2/R3 pedestal divider. Total G=139.2	forced by the AD8226 input-voltage-range inequalities (Rev.C p.20 Eq.1-3 + Table 8, validated against published Fig.12/13 before use): at Vcm=1.65 V on a 3.3 V rail the amplified differential swing at the internal gain-stage nodes is capped at 1.90 V (25 C) and 1.33 V in the rail-5pct/excitation+5pct/0 C corner, so max single-stage gain is 66-95, not 147. The datasheet prescribes exactly this remedy. Returning the stage-2 gain resistor to /VREF makes the output pedestal EXACTLY Vref with no matched network and cancels reference drift to first order
P2	Electrical specs derived away from the brief (NOT mode relaxations): gain 165 -> 139.2; pedestal ~0.1 V -> 0.252 V; usable output window 0.05/3.25 V -> 0.10/3.20 V; transfer Vout = 0.252 + 139.2*Vdiff (-1 mV -> 0.113 V, 20 mV -> 3.037 V, +25 mV clips gracefully)	gain must fit under the -5pct rail ceiling of 3.065 V; the -1 mV worst case of Q5 needs 0.139 V of headroom below zero scale plus the stage output floor. A mode relaxes geometry/cost/packaging only, so each of these is recorded as an electrical derivation with its mechanism
P2	ACCURACY SPEC MISSED AND ACCEPTED: Q7's 5 uV RTI over 0-50 C is not met. Offset drift 13.9 uV typ / 56.4 uV A-grade max over +-25 C from a 25 C calibration, plus 30-87 uV of gain drift at full scale. Pedestal additionally tracks the rail at 548 uV RTI per volt	5 uV RTI on a 20 mV full scale over 25 C is 10 ppm/degC of full scale - a laboratory-grade tempco, not a 12-bit one - and no 3.3 V-capable part in the candidate set reaches it. The alternative (INA333 at G~40 plus a zero-drift second stage) spends the entire bandwidth margin (8.75 kHz vs 45 kHz), lands ON the budget typical-only, and leaves the largest term (gain drift) untouched. Accept-and-record chosen per the owner's learning-value instruction; the miss is quantified here and shown at H1, never dropped
P2	Stackup JLC2313_1.6, 2 layers, class 2L	no controlled impedance exists on a DC-1 kHz chain; the input pair needs a CONTINUOUS ground reference (solid B.Cu pour, no slot under the differential run), not a close one. 4L would duplicate that job and distribute 0.65 mA
P2	No output RC filter. Chain noise is 0.88 uVrms RTI band-limited to 1 kHz but 4.97 uVrms (33 uV p-p) over the chain's own 41 kHz	an output filter is filtering the datasheets do not require, which block-only scope excludes. Recorded so the number is visible: the reader must band-limit to get the low figure. This is a scope decision, never a claim that the filter is unnecessary in general
P2	AD8227 (pin-compatible, named by the AD8226 datasheet as the other escape from the Eq.2 boundary) deferred to the P2 research task as a directed sub-question rather than settled by assumption	LCSC has it but thin (AD8227ARMZ-R7 MSOP-8, 201 stock, USD 2.99; the SOIC-8 A-grade has 18). Its boundary constants and its bandwidth at G~140 are unverified, and the architect correctly refused to design against unverified numbers. The researcher has allowlisted analog.com access, so a cited answer costs nothing extra and becomes a knowledge record either way
P2	Mode relaxed NOTHING on this board	no dimension is stated anywhere in the brief, geometry is an output of the placement under the canonical binding, and cost/packaging were untouched

Phase	What	Why
P2	analog.com is unreachable from this host - whole-host timeout, not just a PDF path (research.py fetch ReadTimeout twice; bare curl to https://www.analog.com/ returns 0 bytes after 90 s). Treat as a STANDING condition, not a blip: P1's reference-design agent hit the same thing independently	consequence for this board: no vendor-layout-tier source exists for the actual part (AD8226), so the interface records are all cross-vendor tier witnessed by TI's INA333 instead of ADI's own text. WORKAROUND that works and should be used from here: the AD8226 datasheet is obtainable through the allowlisted LCSC mirror (wmsc.lcsc.com), and the block-inamp-1 task already has 2304140030_Analog-Devices-AD8226ARZ-R7_C34250.pdf in the shared sources dir
P2	The interface:in slot will be RETYPED before the post-research coverage re-run: its operating_point currently carries only diff-pair geometry (gap_mm/max_skew_mm/max_uncoupled_mm/term_ohm) so none of the five records' envelopes can be tested and they can only ever read provisional	measured by the researcher, not assumed: it ran envelope containment against the slot and all five topology records returned unknown. A kilohertz-rate 20 mV differential sensor input is not a controlled-impedance pair - what it needs is, rsource_ohm, vcm_v, vout_v, vsig_mv, dt_c, err_rti_mv, ibias_ua, mismatch_pct. diff_pairs[].operating_point is supported by knowledgelib, so the retype is legal without a schema change. Deferred until the block task closes so the AD8226 input bias current comes from a cited page rather than a guess
P2	AD8227 stays UNCLOSED and the design does NOT switch to it. The two-stage AD8226 chain stands	its datasheet is unacquirable from this host today: analog.com and mouser.com stall research.py fetch, and LCSC's mirror serves a ~7 KB 'Datasheet temporarily unavailable' stub for every ADI part. Neither its input-range constants at Vs=3.3 V / Vcm=1.65 V nor its bandwidth at G~140 were ever read. The derived mechanism points to yes (an in-amp's minimum gain IS its difference-amp gain, so min-gain-5 means one fifth the internal-node swing at equal total gain) but a derivation is not a datasheet, and trading a validated design for an unverifiable part is the wrong trade. Recorded as the highest-value next acquisition, together with ADI AN-1401 'Instrumentation Amplifier Common-Mode Range: The Diamond Plot'
P2	P6/P7 CORRECTION: do NOT spend layout effort on input trace/via length matching and plane-continuity fetishism for CMRR. The rule CMRR=(G+1)/(4t) belongs to a four-discrete-resistor difference amplifier where G and t are the SAME network; an in-amp's ~100 Gohm inputs convert essentially nothing from series-path mismatch at DC-1 kHz with a 350 ohm source	second reader refutation, page-level: every SBOA582 circuit is a four-resistor diff-amp at G=1, and the transfer to stray trace asymmetry was asserted rather than argued. The error term that IS verified at the 5 uV RTI budget is the TEMPERATURE GRADIENT across the two input junctions - so the placement rule that matters is keeping the two input poles at the same temperature and away from anything that dissipates, not matching their lengths
P2	BRING-UP NOTE: a missing input bias-current return path does NOT park the output at a rail - it produces a plausible near-zero output that 'appears normal'	second reader found the researcher's claim contradicted by INA333 s8.2.2.6 on the same cited page and by TI SLYT226 p.1. This matters because the failure mode is silent: a board with a floating input reads like a working board with no signal. Belongs in the bring-up notes

Phase	What	Why
P2	interface-in-1 ledger quality: 1 of 6 records verified. The refutations are themselves the finding - the whole ledger is TI/Microchip-witnessed and NO number for this board's actual part (AD8226) is witnessed anywhere in it, so the bias-return REQUIREMENT transfers but the SIZING does not	analog.com is unreachable (standing condition), so the interface researcher had no vendor witness for the board's own part. The block-inamp-1 task has since proved the AD8226 datasheet IS obtainable from an allowlisted Farnell mirror - that source did not exist when this task ran
P2	The architecture's foundation is VERIFIED on the page and the generalization is stronger than the architect claimed: AD8226 printed p.19 gives Vout = G(Vin+ - Vin-) + Vref exactly, so Eq.2 becomes Vcm + |Vout - Vref|/2 < Vs - V+LIMIT with NO GAIN TERM IN IT. Confirmed on the printed figures, not just algebraically: Fig.9 (G=1) and Fig.12 (G=100) at Vs=+2.7 V differ by at most 0.1 V at four vertices, and the Vref=0 diagonal from (+0.02,+2.0) to (+2.4,+0.8) is the SAME straight line in both - that line IS Eq.2. All twelve Table 8 constants exact	the two-stage topology is downstream of this claim, so it was targeted for refutation by a fresh reader and survived. The practical restatement: the in-amp's constraint is on ITS OWN OUTPUT SWING, not on its gain - which is exactly why moving gain to the second stage relieves it
P2	AMENDING the earlier P6/P7 symmetry ruling. Do NOT chase input trace LENGTH matching for DC CMRR (that stands - an in-amp's ~100 Gohm inputs convert nothing from series mismatch at DC). But DO keep the two input paths electrically similar - same width, same neighbourhood, same environment - because AC CMRR does degrade with SOURCE IMBALANCE	the block second reader refuted the parasitics record precisely here: its envelope claimed no vendor number exists above 5 kHz, but the ledger's own AD8226 PDF p.13 carries Fig.27 'CMRR vs Frequency, RTI' and Fig.28 'CMRR vs Frequency, RTI, 1 kohm Source Imbalance' out to 100 kHz, and Fig.28 measures this exact mechanism. Our chain is open to 41 kHz and common-mode interference rides in there. Our own imbalance is far below Fig.28's 1 kohm case, so this is cheap insurance, not a driver - the dominant term at the 5 uV budget remains the thermal gradient across the two input junctions
P2	AD8226 A-grade input bias current, cited from Table 3 printed p.5: 20 nA typ / 27 nA max at +25 C (5 nA min), 30/35 nA at -40 C, 15/25 nA at +125 C, avg TC 70 pA/degC. Input offset current 1 nA max. The input stage is pnp, so bias current always flows OUT of the part - the return path must SINK it	needed to declare the input interface's operating point without guessing; obtained by asking the second reader while it had Table 3 open rather than by inventing a number. The pnp direction is a bring-up-relevant detail: the return network sinks current, it does not source it

Phase	What	Why
P2	P6/P7 LAYOUT RULE, now witnessed by the board's own part (AD8226 Fig.28): SOURCE-PATH IMBALANCE, not gain, sets CMRR here. Balanced (Fig.27) LF CMRR climbs 96/117/133/144 dB across G=1/10/100/1000; with a 1 kohm source imbalance it climbs only 96/102/106/108 dB, and above ~1 kHz the G=1/10/100 curves collapse onto one roll-off reaching 55-68 dB at 100 kHz. At this board's G1=39.9 that imbalance costs about 25 dB at the 1 kHz signal frequency	supersedes the vaguer earlier ruling with a measured number from the actual part. Consequence for this board: it adds NO series component in either input leg, and a 350 ohm bridge's arm-to-arm imbalance is orders below Fig.28's 1 kohm case, so we are far from the penalty - but the rule to enforce at P6/P7 is that nothing may be inserted in one input leg and not the other, and the two paths must see the same environment. That is a different instruction from length-matching, which still buys nothing at DC
P2	Recorded as a research-quality practice worth keeping: the researcher caught that AD8226 Figures 27/28 are taken at Vs=+-15 V (TPC conditions header, printed p.9) while Table 3 is at +Vs=2.7 V single supply, so the absolute dB from those curves cannot be set against Table 3 minimums - only the PENALTY (the ratio of two curves under identical conditions) transfers	without that check the record would have carried a false claim that source imbalance drops the part below its guaranteed CMRR spec. Both records state the limitation explicitly. This is the failure mode adversarial review exists to catch, caught by the author instead
P2	CORRECTION to the P6/P7 CMRR layout rule: the 1 kHz source-imbalance penalty is about 16-17 dB (17.0/15.9/16.3 dB at G=10/100/1000), NOT the ~25 dB I recorded earlier. The qualitative rule is unchanged - source-path imbalance, not gain, sets CMRR here - only the magnitude was overstated	second-reader refutation with page-level measurements: at 1 kHz Fig.27 reads 109.4/109.4/110.6 dB and Fig.28 reads 92.4/93.5/94.3 dB. The researcher's claimed 27/23/19 dB was 3-10 dB out, outside its own stated +-2 dB curve-read tolerance, and that is exactly the number the rule hands a designer at this board's 1 kHz signal frequency. The reader also verified everything else in the record (plateau ladders within 1 dB, gain-to-curve mapping via the Fig.28 arrow callouts, the +-15 V TPC conditions caveat, envelope bounds) and found a bonus fact: Table 2 (printed p.3) is the matching +-15 V table carrying the SAME minimums as Table 3, so the conditions caveat costs nothing
P2	CORRECTION to the earlier bring-up note. A missing bias-current return path has NO diagnostic symptom: the AD8226 page states none; Renesas AN1298 p.14 says the output will slowly drift until it saturates into a rail; the same vendor's sensor-health paper says a saturated input stage can appear normal to the processing circuitry. Both endings are real. The honest bring-up rule is: PROBE THE INPUT PINS, NOT THE OUTPUT	my earlier note recorded only one of the two endings (plausible near-zero output), inheriting it from round one's TI-witnessed record. The re-witnessing pass found the contradiction is between two real documented behaviours, not an error in either, so the diagnostic value is in the input pins' DC level

Phase	What	Why
P2	Bias-return sizing now witnessed by the board's own part, and it confirms this board needs NO bias resistors: the 350 ohm bridge is excited from a 3.3 V supply sharing our ground, so each input already has a DC return through its own bridge arms and the J1 GND pole. The sizing rule matters only if the source were floating - and it is brutal there: AD8226's own printed value is 10 Mohm (single resistor, Fig.62), but with a single resistor the far input's bias current returns through the 350 ohm source for about 10 uV differential, already past the 5 uV budget. A balanced pair caps R near 3.7 kohm (differential term = Ios*R + Ib*R*mismatch), which then loads a 350 ohm source by about 4.5 pct. Round one's transplanted 47k figure would have cost about 60 uV differential on this part	this is exactly the claim the first reader said would not transfer - the REQUIREMENT transfers, the SIZING does not - and re-witnessing proved it right: the CMOS 47k/10k values were off by more than an order of magnitude for a 27 nA pnp-input part
P2	P6/P7 input-path rule sharpened to a number: with 0.8 Gohm||2 pF inputs the DC resistance term is dead, and it is CAPACITANCE match that matters - 1 pF of path mismatch gives -119 dB at 1 kHz, 10 pF gives -85 dB at 5 kHz against the part's own 90 dB spec	gives placement and routing a checkable target (keep the two input paths' capacitance to the pour matched to about 1 pF) instead of the untestable instruction to match lengths. Consistent with, and more actionable than, the Fig.28 source-imbalance rule
P2	P2 exit coverage passes: exit 0, 2 slots, 0 gaps, all 11 classes provisional. No 'designing under coverage gap' decision is needed	16 verified records and 2 new checklists (inamp, in) now cover both slots. Provisional is the correct ceiling here and not a shortfall: research narrows a gap to provisional by contract, the owner's promotion ruling makes a record approved, and only bring-up evidence makes it proven
P2	Research spend: 4 tasks over 2 rounds, all 6 of the per-run budget. Round 1 (13 records) was witnessed by the wrong parts because analog.com was unreachable - 7 refuted. Round 2 (9 records) re-witnessed from the AD8226 itself via an allowlisted Farnell mirror - 9 verified, 9 of 9	the second round was worth its cost and the evidence says so: it closed all five interface gap classes, replaced a transplanted 47k bias-return figure that would have cost ~60 uV differential on this part, and produced a corrected CMRR-vs-imbalance number that a layout rule now rests on. Deciding to spend it rather than record 'designing under coverage gap' was the right call for a board whose stated purpose is to seed the library

Phase	What	Why
P3	Connectors J1/J2/J3 are hand-soldered THT (KF128-5.08 family, one-piece board-side body); everything else is JLC-placeable SMT on single-sided top assembly. J1 is a single 3-pole body	JLC's standard SMT service does not place screw terminals, which requirements section 7 anticipated. The single 3-pole body is not a convenience: it is what makes the two input poles' thermocouple junctions identically built, which is the one error term the research leg verified as dominant at the 5 uV budget. Pitch 5.08 mm chosen over the research's 3.5 mm read because blocks.md names 5.08 three times and the larger THT pads suit hand-soldering
P3	Sourcing risk recorded: the AD8226's only pin- and gain-law-compatible alternate (AD8226BRZ-R7) has 17 pcs in stock - usable once at qty 5, not resupply-safe. AD8227 was rejected as the alternate because its internal gain law differs and would force an RG recalculation. Separately, KF128-5.08's wire window is 22-12 AWG, narrower at the fine end than the 24-16 AWG requirements assumed	both are honest flags rather than blockers at qty 5. The AWG one is not resolvable from this workspace - it depends on the owner's actual sensor cable - so it is recorded for confirmation before ordering rather than guessed
P3	ADDING a series output isolation resistor (R6) between U2B's output and J2, overruling blocks.md B5's 'no series isolation resistor needed'. Value is NOT fixed here: P4 must justify it against OPA2333 Figure 15 and it must be verified by the B4 step bench run with CL=1 nF	the datasheet extraction found Figure 15 shows overshoot climbing to roughly the mid-30 percent range by a 1 nF load, and requirements 9a Q8 explicitly allows up to 1 nF of output cable. That is a stability problem at the STATED operating point, so the resistor is datasheet-required support, NOT excluded 'filtering' - block-only excludes conditioning the datasheet does not require, and this one it does. Cost is one 0603 and a ˜0.1 pct gain error into a >100 kohm load, which calibrates out. Recorded as an overrule of a P2 ruling with the evidence, not a silent edit
P3	U2A's REF-drive impedance is a SIM item, not a datasheet item: TI gives open-loop output impedance 2 kohm typ but only at f=350 kHz, and states no DC or closed-loop figure. The AD8226 needs its REF driven below 2 ohm	with AOL 130 dB typ a unity-gain buffer will deliver well under 2 ohm at DC, but that is an inference, not a vendor number. Added to the P4 sim brief so a bench proves it rather than an argument asserting it
P3	OPA2333 output swing CONFIRMS the architecture at the worst corner: guaranteed 70 mV max to each rail over temperature at RL=10 k, valid across the whole 1.8-5.5 V range, so at 3.135 V the floor is <=0.070 V and the ceiling >=3.065 V. The design's 0.113 V and 3.037 V clear by 43 mV and 28 mV, with the real >100 kohm load much lighter than the 10 k test condition	this was the open question I put to the extractor, because in this topology it is the op-amp and not the in-amp that faces the rails. It holds

Phase	What	Why
P3	P3 exit coverage passes: exit 0, 4 slots, 0 gaps - the two PART slots (U1 inamp, U2 ref_gain2) read COVERED off the datasheet extractions, and the two block/interface slots read provisional. No further research tasks opened, with 2 of the per-run budget left unspent	the seeding-floor call (–maturity-floor proven –research-provisional) re-reported the block and interface slots as gaps, but that is the escalation doing its job, not new ignorance: those two slots were already researched across two rounds to 16 verified records, and research CANNOT raise maturity past verified - draft-to-verified is the second reader's verb, verified-to-approved is the owner's, approved-to-proven is bring-up's. Opening a third round would spend budget to re-derive what is already on disk. build-modes.md prescribes exactly this: run the first call of each phase escalated, then re-run WITHOUT the flags to fold in what was learned, because re-escalating only re-fires the trigger
P3	Both datasheet extractions validate: 8 pins each, land patterns present, 10 layout_notes each - well clear of the thin-extraction gap threshold of 2	the layout notes are what P6/P7 are held to, and a thin section is a detectable coverage gap by design. These are substantive: AD8226's own layout section plus Figure 61, and OPA2333's decoupling/thermal-EMF/auto-zero guidance
P3	SIM BRIEF CORRECTION for P4/P8: bench B1's 'INL <= 0.05 pct FS' window is NOT a datasheet-derived bound. The AD8226's single-supply Table 3 has no Gain Nonlinearity row at all - it exists only in the dual-supply Table 2. Either drop the INL bound, or keep it explicitly as a DESIGN TARGET labelled as such in the bounds sidecar	the sim gate reads the bounds sidecar as truth, so an invented bound dressed as a vendor number is exactly the defect the gate exists to catch. Found by the extractor cross-checking the architecture's sim windows against the tables
P3	Recorded as interpolation, not vendor data: this board runs at G1=39.9, which is not a tabulated point - the AD8226 datasheet gives CMRR and bandwidth only at G=1/10/100/1000. blocks.md's ~45 kHz / ~1.8 MHz GBW figure is an interpolation between table points	the second reader already verified the interpolation is legitimate (the tabulated points are exactly constant-GBW and Fig.26 is a clean single-pole family with no peaking), so this is a provenance label rather than a doubt. It matters because the 41 kHz f-3dB in constraints.json inherits it
P3	Two further single-supply table limits noted: AD8226 Table 3 characterizes output swing only at RL=10 kohm (no single-supply 100 k row; the dual-supply table's 100 k row is numerically identical), and input offset current at -40 C reads 1 nA in Table 3 versus 2 nA in Table 2 - a genuine table-to-table difference, not a typo. Table 3 is primary because it is our condition	both are cases where the honest answer is 'the vendor did not characterize our exact condition'. Neither changes the design - the 10 k swing figure is conservative for our >100 kohm load, and 1 vs 2 nA of Ios is far inside the bias-return headroom - but both are now on the record rather than silently smoothed

Phase	What	Why
P3	J2/J3 footprint drill corrected 1.6 mm -> 1.4 mm to match the KF128-5.08 datasheet and J1's already-exact 3-pole pull. The two LCSC catalog entries for one connector family disagreed with each other	1.6 mm is not unusable (0.40 mm annulus, far above the 0.15 mm floor) but it is off-datasheet, inconsistent with J1 on the same board, and adds a second drill size for nothing. On IPC grounds 1.4 mm is lead+0.20 for the ˜1.20 mm pin diagonal - the normal window - while 1.6 mm is lead+0.40, loose. These three connectors are the only parts that take mechanical load (wires torqued into screw clamps), so a loose barrel is the wrong way to err. Fix routed back to the librarian rather than done by the orchestrator: a .kicad_mod is a design file and rule 1 forbids opening one
P3	Accepting the U2 SOP-8 pad-size warning (0.63 x 1.865 mm pulled vs TI's recommended 0.6 x 1.55 mm) rather than forcing the vendor land pattern	the pulled footprint is JLC's own catalog geometry, and JLC is the assembler - their pads are what the stencil and placement machine will actually use. Consistency with the assembler beats vendor-exactness for a part being machine-placed
P3	U1 SOIC-8 pad size is UNVERIFIED and recorded as an inherited gap: the AD8226 datasheet contains no recommended land-pattern drawing at all, only the JEDEC MS-012-AA package outline, so fp_verify had no number to compare against	honest gap rather than a false pass. The 0.568 x 1.95 mm pull is a standard narrow-body SOIC-8 for the standard MS-012-AA body, so there is no basis to call it wrong - but there is also no vendor drawing behind it, and the record should say so
P4	ONE schematic agent for the whole board, not one per sheet	sheets.md rules one sheet and no hierarchy: 15 parts on a single signal path. The SKILL's lean amendment permits one schematic agent on a small board provided it is recorded - this is that record. It also keeps /VREF visible as one node touching U1 pin 6, U2A's output and the stage-2 gain return, which is the thing a reader of this board should see at a glance
P4	NO mounting holes on this board. Decided now, before P5, because adding any part after board_init discards placement and routing	block-only scope excludes mechanical, requirements section 5 says none are required for bench use, and constraints.json names none. Section 5 did leave the door open to M3 clearance holes if the earned outline had corner room - I am closing that door deliberately rather than letting it become a late surprise, since the geometry is an OUTPUT here and the outline will be shrunk to the placement at P6
P4	R6 sits OUTSIDE U2B's feedback loop: R4's feedback is taken from /AMP2_OUT (U2.7) before R6, and /VOUT is R6.2 to J2.1	correct for a series isolation resistor - inside the loop it would not isolate the capacitive load at all, and the loop would fight the cable. Outside the loop it costs a divider error against the load, which is 0.1 pct into the specified >100 kohm and calibrates out. The schematic agent drew it this way independently

Phase	What	Why
P4	REVERSING my P3 decision: R6 is REMOVED. The premise I added it on was wrong. OPA2333 Figure 15's ˜mid-30 pct overshoot at 1 nF is a UNITY-GAIN figure, and U2B runs at noise gain 3.49. The same macromodel that reproduces Figure 15 exactly (32 pct at 1 nF, unity gain) gives 6.6 pct for the as-built stage with NO R6 - about 63 deg phase margin, 50 deg on the pessimistic ro=2k bracket	at block-only scope, conditioning the datasheet does not require is excluded by the tier, and the bench has now shown it is not required. 100 R would not have done the job in any case: an out-of-loop isolation resistor has authority R6/ro, so against ro=1-2 kohm it buys only 0.3-1.3 points of overshoot (6.60->6.28 pct calibrated, 17.94->16.68 pct pessimistic). The value that would actually isolate is 470 R or more, but that drops the -5 pct-rail full scale from 3.022 V to ˜3.011 V, under blocks.md B6's own 3.02 V figure. Removing pre-P5 is the cheapest moment - after board_init it would be a remove-part operation. The sim-analyst recommended keeping it as harmless; I am overruling that because an unjustified part on a bare-bones board is exactly what the scope tier forbids, and because the removal is the more useful lesson
P4	U2A REF-drive impedance measured, and the <2 ohm rule is EXCEEDED above 700 Hz - assessed as NOT a defect. Closed-loop Zout: 0.43 mohm DC, 0.17 ohm at 60 Hz, 2.86 ohm at 1 kHz, 116 ohm at 41 kHz (pessimistic ro=2k: crosses at 350 Hz)	the AD8226's 2 ohm figure sits in its datasheet's DC-to-60 Hz CMRR context; this board's common mode is a STATIC 1.65 V per Q1 with no 1 kHz common-mode content; and the residual CM gain error Rref/100k works out to 90.8 dB (84.9 dB pessimistic), at or above the AD8226's own 90 dB minimum at 5 kHz. Nothing on the board could flatten it anyway - C4 is upstream of the buffer, and a cap on the buffer output is the follower-into-capacitance blocks.md s5.5 forbids. Gated as warnings with that reasoning in the sidecar rather than hidden
P4	The design point is CONFIRMED by simulation against the exported netlist, not against the architecture's arithmetic: slope 139.096 V/V, Vout(0)=0.2528 V, Vout(20 mV)=3.0347 V, Vout(-1 mV)=0.1137 V, clip 3.2796 V monotonic. Stage-1 output at -1 mV sits at 0.2123 V against the AD8226's 0.10 V guaranteed floor - the P1 reconciliation (Ruling 1) holds with 112 mV to spare	the analyst diffed its hand-typed topology against kicad/bb-amp.net when the netlist appeared mid-run (sim_run –fragment): every value and every connection matched, including R5 returning to /VREF. That is real wrong-value verification rather than a restatement of the design intent
P4	R6 removal folded into the benches and the board measures BETTER without it: b1 slope 139.096 -> 139.235 (exactly the design, no divider), full scale at the -5 pct rail 3.0221 -> 3.02513 V so blocks.md B6's 3.02 V figure is met by 5.1 mV instead of 2.1 mV, and delivered span 99.874 -> 99.974 pct. Shipping stage into the specified 1 nF: 6.60 pct overshoot (˜63 deg phase margin), 17.94 pct on the pessimistic ro=2k bracket (˜50 deg)	blocks.md B5's original objection to any series resistor is now answered on measurements rather than traded away, and the stability claim the board rests on is gated on both ends of the ro bracket

Phase	What	Why
P4	The R6 removal exposed TWO stale constants in the benches themselves, both caught by the sidecars: b4's settling thresholds were pinned to the old R6-loaded final value (3.03747) so t_rec stopped crossing and returned as a MISSING measure, and b3's f-3dB trigger was pinned to 20*log10(139.10)-3. Both retuned	worth recording as evidence the sim gate works on its own author: a bench that silently carries a stale constant is exactly the failure mode a bounds sidecar exists to catch, and here it caught the analyst rather than the board
P4	W1 ACCEPTED as a real gap in the recorded accuracy budget: blocks.md Ruling 3's error table counts the R2/R3 ratio TCR (1.1 uV) but OMITS the rail term that ratio multiplies. The pedestal is 7.63 pct of the rail and reaches the output at gain 1, so dVout/dVs = 0.0763 (measured by bench B6), and 0.28 pct of post-calibration rail movement - about 9.1 mV on 3.3 V - consumes the entire 5 uV budget on its own	the fact itself was recorded at P2 ('pedestal tracks the rail at 548 uV RTI per volt'), but it never made it into the error-budget TABLE that H1 presented, which is the artifact a reader trusts. Not an electrical error and not a scope violation - a reference IC is out of scope at block-only and a static pedestal calibrates out - but on a teaching board an incomplete budget table is a defect. The reviewer is right and the table must carry the term
P4	W2 and W3 are WRONG STATEMENTS ON A TEACHING BOARD and will be fixed, not waived. W2: the sheet claims U2A buffers the pedestal to '<2 ohm at U1 REF' unqualified, while this board's own B9 bench puts the 2-ohm crossing at 350-700 Hz and reads 2.86-5.71 ohm at 1 kHz. W3: the equations box attributes 'Eq.2 allows 1.33 V, G<=66' to Vcm=1.65 V, but Eq.2 at 1.65 V gives 1.49 V and G<=75 - the 1.33 V figure belongs to the unnamed Vcm=1.7325 V excitation-+5 pct corner	this board exists to be studied, so a wrong equation or an unqualified claim in its own documentation is a real defect with a real victim - the next reader. Both are cheap to correct and neither touches the netlist
P4	W4 is a latent landmine and will be fixed in the library: KF128-5.08-2P carries reference prefix 'U' while its 3P sibling carries 'J'. A re-annotate-with-reset would rename J2/J3 and silently break constraints.json, parts.json and the P9 CPL	harmless today because the refdes are already assigned, which is exactly why it would go unnoticed until something re-annotates. Cheap to fix now, expensive to diagnose later

Phase	What	Why
P5	Declared impedance_ohm: 150 on the /IN_P + /IN_N pair, with an in-file _impedance_note saying it is NOT a controlled-impedance target. 150 ohm is what this pair ACTUALLY MEASURES at the 0.3089 mm width we want on this stackup (h=1.53 mm, er=4.5, 1 oz) - a consequence, not a requirement. Result: 0.3089 mm width, 0.3 mm gap, netclass Diff150	rules_gen has no way to express 'route these symmetrically, no impedance target', and its 90 ohm default would have produced a 1.3743 mm pair. Removing the diff_pairs entry instead was rejected: knowledge retrieval keys the interface records off diff_pairs[].base, so deleting it would have silently dropped all eight verified interface records - thermal EMF, aggressor separation, path equivalence - out of the P6 and P7 spawn prompts, which is where they are actually needed. Declaring the true measured impedance is the least-bad lever and it states a fact rather than inventing a target
P5	P5 setup complete: board_init self-check parity 0, setup_violations 0, clean true (30 unconnected are P7's). Stackup JLC2313_1.6, 2 layers, fab profile 2layer_1oz with floors min_clearance 0.127 / min_track 0.127 / min_via 0.6 / min_hole 0.3 / annular 0.15 / hole-to-hole 0.5 / copper-to-edge 0.3, all at ERROR severity. 14 components, 11 nets, 0 mounting holes. Provisional outline 41.0 x 57.7 mm	the provisional outline is deliberately generous and is NOT the board: under the canonical binding geometry is an OUTPUT, so P6 places to the canonical layout and board_edit –outline fit earns the real size afterwards. Reporting the fab floors explicitly because a clean DRC against floors below the fab minimums means nothing - that has shipped twice on prior boards
P6	CORRECTION to my own P5 record: board_init.json.outline_bbox is [13.0, 13.0, 41.02, 57.69] = MIN/MAX CORNERS, not a size. The provisional outline was 28.02 x 44.69 mm, not the 41.0 x 57.7 I recorded in the P5 digest and passed to the placement brief	caught by the placement agent, which then found the canonical layout needs ˜46 mm of width and could not fit in 28 mm at any density. It grew the room deliberately, placed to the layout, then earned the size back. The order is load-bearing and is now a workspace learning: –outline fit CLIPS content to the current outline, so placing outside the room first and fitting afterwards silently returns a smaller board
P6	Board size EARNED by the placement: 47.98 x 28.251 mm (content 45.980 x 26.250 at a 1.0 mm margin). Gate place PASS, 0 failing across all five legality legs; DRC 30 findings all unconnected, 0 copper/silk/courtyard/parity; check_decoupling pass; route probe on a scratch copy completion 1.00, 0 unrouted of 30	geometry is an OUTPUT under the canonical binding and this is what the layout needed - no dimension was ever treated as a target. hpwl fell 323.181 -> 173.45 mm from seed to final. The annealer's four candidates were all REJECTED and the canonical tile hand-built instead, because with movable_clusters 2 of 5 every candidate HPWL was dominated by connectors pinned to a provisional outline - stage 2 was noise on this board and the agent said so up front

Phase	What	Why
P6	TAKING the J1 pin-swap fix: J1 pole 1 becomes IN-, pole 2 becomes IN+. J1's poles run (IN+,IN-,GND) while the AD8226's inputs run (-IN,RG,RG,+IN), so the pair must cross exactly once under ANY rotation or edge assignment - verified across all eight combinations, so it is structural, not a placement artifact	the crossing is NOT a defect (2 vias on one leg, ˜0.6 pF imbalance, -123 dB at 1 kHz, inside the declared 1 pF budget and 33 dB under the part's own 90 dB spec). But the fix is a LABEL SWAP with zero electrical consequence that buys a fully planar mirror-symmetric zero-via pair, and input-path symmetry is the single thing this board exists to teach. Two vias would also punch the B.Cu reference directly under the input pair and add asymmetric junctions on the net whose thermal-EMF symmetry is the dominant 5 uV-budget term. board_update REFUSES pad-net rewires by design, so this costs a P5 re-init - but the ops trail (kicad/canonical2.ops.json) makes the replace deterministic rather than a true rewind
P6	J1 pin swap VERIFIED to deliver: input pair HPWL 15.23 -> 11.43 mm per leg, board HPWL 173.45 -> 165.85, signal crossings 6 -> 5 with the pair crossing gone, length delta 0.000000 mm, mirror axis y=30.0000 exact (= U1 centre = R1 centre). Proven by a machine test, not an argument: two straight F.Cu segments laid on a scratch copy with ZERO vias gave 0 non-unconnected DRC findings and dropped unconnected 30 -> 28. Outline reproduced identically at 47.98 x 28.251 mm - fit is idempotent to the micron	the replay had to prove both that the layout reproduced and that the swap bought what it was for. The route probe alone could not have answered the second question - it measures completion without importing copper, so it cannot say whether the pair took vias. The two-segment DRC test is what actually settles it
P6	CORRECTION to my own replay instruction: the grow step cannot come first. Growing the outline against a fresh shelf pack REFUSES (10 blocking items - 2 footprint_outside, 8 copper_to_edge, because the shelf puts U1 and J3 below y=53). The real order is PLACE -> GROW -> FIT	I wrote 'open the room first, then place, then fit' in the handoff. The agent had reported the principle, not the literal step order, and re-derived the correct sequence rather than following my wrong instruction. The invariant that actually matters is that runs only when every item is already inside the outline, because fit CLIPS
P6	ACCEPTING the silk residual: C2's and C4's refdes land under J3's and U2's package bodies - legible on a bare board, hidden once assembled. silk_place reports 0 residual at 0.30 mm clearance and DRC finds 0 silk issues	the alternatives both cost more than the defect: clearing them needs ˜1.5 mm of extra board height, which fights the earned-outline principle on a board whose binding makes size an output, or moves C4 off the /VREF_SET node it filters. A refdes under a body is normal and cosmetic; a filter cap moved off its node is not
P6	Residual imbalance recorded, not actionable: J1 pole 3 (GND, the bias-return path) is adjacent to whichever signal pole sits in the middle, which after the swap is /IN_P - its corridor is 3.726 mm from the GND pad where /IN_N's is 8.719 mm. Same magnitude as before the swap, opposite leg	this is the connector's own 3-pole order, not the layout's, and it cannot be removed without a different connector. Both input pads still have identical pour clearance rings, neither takes a pour tie or thermal relief, nothing within 10 mm dissipates, and the vertical plate capacitance that dominates the match is identical. Recorded so it is visible rather than discovered later

Phase	What	Why
P6	Amending the prior record, whose why-text lost a word to shell substitution: the invariant is that board_edit –outline fit runs ONLY when every item is already inside the outline, because fit CLIPS content rather than growing to hold it	restating it cleanly so the record is not missing the operative verb. The step order is place -> grow -> fit; growing against a fresh shelf pack refuses because items sit outside the provisional room
P7	MY ERROR in the P7 brief: I cited in-aggressor-separation as a governing record, but that record was REFUTED by the round-1 second reader and is still draft. The selector was RIGHT to exclude it - drafts never inject, and 22 ledger ids minus the 6 refuted equals the 16 that were handed over. The router flagged the discrepancy, read the draft directly and honored it anyway	the separation instruction was still sound and P6 had already provided it (J1 to J2 35.82 mm with the pour between), so no harm reached the board - but I presented a refuted claim as settled knowledge, which is the exact failure the draft/verified distinction exists to prevent. The honest support for keeping the output away from the input is the round-2 verified in-input-path-equivalence record, not this one
P7	MY ERROR in the P7 brief: I named /AMP2_OUT as a net to keep clear. That net stopped existing when R6 was removed at P4 - the real nets are /AMP1_OUT (U1 out, 39.9 V/V) and /VOUT (U2B out, 139 V/V). The router caught it and measured both as aggressors	stale net name carried forward from the pre-R6 topology in my own head. Recorded because it is the second time this run that a downstream brief of mine referenced a superseded artifact, and the pattern is worth seeing
P7	Router OVERRULED route_critical –only diff on the input pair, and was right to. That path drives route_diff.py which REQUIRES the pair to emerge as a coupled pair at 0.3 mm gap and errors otherwise - it would have converged both legs onto y=30.0 straight into R1's pads, forcing bends and a detour. It laid two straight route_edit segments instead, geometry pre-flighted, and Freerouting then preserved them byte-exact as protected guide wires	third independent sighting of the same tooling gap: diff_pairs conflates 'route these symmetrically' with 'control the impedance of these'. It bit at P5 as a 1.3743 mm netclass width, at P2 as a coverage slot typed in geometric dims, and now at P7 as a coupling requirement that would have destroyed the symmetry it was meant to protect. The router cited in-constraints-emission (pair_rule_is_equality) and constraints.json's own _impedance_note to justify the overrule
P7	Two measured items accepted rather than chased: (1) Freerouting's /RG_A and /RG_B escapes are not exact mirrors - 13.3 um of difference in their distance to the input legs, bounded arithmetically at <=0.003 pF or about -170 dB of CM-to-DM conversion at 1 kHz, roughly 80 dB below the part's own spec. (2) Tightest clearance on the board is /AMP1_OUT to /VREF at 0.1578 mm, legal with 31 um over the 0.127 fab floor	(1) fixing it means ripping and re-laying /RG_B on a board that currently gates clean, for a term 80 dB under the part. (2) /VREF is buffered by U2A to well under 2 ohm at DC, so it is a LOW-IMPEDANCE victim, and the adjacency is forced by U1's own pin order (REF=6, OUT=7 at 1.27 mm pitch) - it is placement, not routing, and the placement is otherwise correct

Phase	What	Why
P7	route_cleanup ran clean and the V13 dangling-pass defect did NOT recur: dry-run proposed 0 ops, live run confirmed drc_before == drc_after with 0 ops applied. The V13 signature would have appeared exactly here - 7 GND stubs whose connectivity runs THROUGH the pour via a terminating via - and it correctly flagged none of them	the playbook asks for dry-run plus inspection on route_cleanup's first live run and says the caveat can be retired once it behaves. On this board it behaved, on the precise case that used to break it
P8	check_diffpair error resolved by FIXING THE PREMISE, not by waiving the finding: max_uncoupled_mm raised 5.0 -> 12.0 (past the 10.81 mm run) with an in-file _coupling_note stating this pair has NO coupling requirement. Verify gate then PASSES, 0 failing	the check demanded both legs run within 0.75 mm of each other and flagged all 10.81 mm as uncoupled. But this pair is DELIBERATELY two mirror-symmetric legs straddling U1's RG pins - converging them to couple would drive both into R1's pads and destroy the symmetry the whole board is built around, which is exactly why the router refused route_critical –only diff. A waiver would have excused a symptom; declaring max_uncoupled_mm honestly states what is true about the board. The equality that DOES matter is independently verified: both legs 10.808963 mm, delta 0.000000, 0 vias, mirror-identical neighbourhood
P8	VERIFY-REVIEWER ERROR 1 CONFIRMED AND BEING FIXED: the board carries 14 pieces of F.SilkS text and ALL 14 are refdes - seven hand-wired screw-terminal poles with no legend at all. The schematic instructs it twice and it never reached the board; the only pole-1 indicator is a 0.21 mm dot under the connector body	these three terminals are hand-soldered and hand-wired by the user, so the silk IS the wiring instruction. J3 reversed puts -3.3 V on U1 pin 8 and U2 pin 8 and destroys both ICs. J1 poles 1/2 reversed pins the output at the bottom rail, which reads as a dead amplifier rather than a wiring error and burns a bring-up session. This is the highest-consequence defect found on the board and it is invisible to every gate - no DRC, ERC, verify or DFM check looks for a missing legend
P8	VERIFY-REVIEWER ERROR 2 FIXED: every human-facing document stated the OPPOSITE pole order. requirements.md sections 2 and 9a Q2, architecture/blocks.md B1 and its block diagram, and the H1 design-doc PDF all said IN+ / IN- / GND. Only the schematic note and the P6 digest were right. Corrected requirements.md (banner at section 2 plus the 9a Q2 row) and blocks.md (both statements); the design doc regenerates from state	the P6 pin swap propagated into the netlist, the schematic and the P6 digest but not into the upstream documents that a human actually reads. A board with no legend AND a document that says the wrong thing is a miswiring waiting to happen. Recorded as a process lesson: a late pin change must sweep every artifact that states the pin order, not just the ones the gates read

Phase	What	Why
P8	Silk legends FIXED: all 7 connector poles now labelled (J1 IN-/IN+/GND, J2 OUT/GND, J3 +3V3/GND), placed outside every connector housing so they survive assembly. Board NOT grown - all 7 fit inside 47.98 x 28.251 mm. Two refdes (R1, C3) nudged ~1.9 mm where they occupied the only viable legend position. DRC 0 violations, verify_all 8/8 checks ran and 8/8 passed	the fixer drove its geometry search through the pipeline OWN parsers (placelib, geom, and check_silk own text-bbox formula) rather than an eyeballed estimate, so its clearance model matches what the gates actually enforce, and it re-derived all seven pole assignments from the netlist before placing a single label rather than trusting my brief
P8	Gates refreshed after the silk edit: place PASS, drc_routed PASS (committed), verify PASS. The place gate had gone stale the moment the board changed, which is what surfaced the report_gen layout-section problem	a silk edit is still a board edit - every gate keyed to the board hash goes stale, and re-running them is the difference between evidence and a memory of evidence
P9	P9 DFM complete: gate dfm PASS, 0 violations at any severity, with the polarity oracle (schematic) and the BOM leg (parts.json) both supplied so it did not pass vacuously. Fab package written: gerbers zip, drill, BOM.csv, BOM-full.csv, CPL.csv	assembly split is correct and checked: 11 SMT parts in CPL.csv, and J1/J2/J3 absent from both CPL and the assembler BOM because they are hand_install - they appear only in BOM-full.csv with their instructions. U1/U2 rotations came through as 180 deg against 270 in the raw position file, which is the jlc_rotations correction the recipe describes as the catcher for a part mounted backwards
P9	RUN STOPS AT P9. No ordering, no H5, no money spent	block-only scope default: 'Stop at P9 (fab package + DFM). Ordering is a separate owner decision.' The order recipe is deliberately not entered - the creation latch and H5 payment step stay untouched, and the release attestation is not built. Everything needed to order exists in boards/bb-amp/fab/ whenever the owner decides to

Sheet plan

bb-amp - schematic sheet plan (P2)

Ruling: ONE sheet. No hierarchy.

The whole board is 14 components on one signal path (J1 -> U1 -> U2B -> J2) plus a two-resistor reference and four capacitors. A hierarchical sheet buys navigation at the cost of hierarchical pins, sheet instances and a netlist whose names carry sheet paths - all of it overhead against a schematic that fits on one A4 page with room for the design equations in a text box. Splitting it would also hide the one thing a reader of this board should see at a glance: that `/VREF` is a single node touching U1 pin 6, U2A output and the stage-2 gain return, which is what makes the pedestal exact (`blocks.md` section 2).

Root sheet = the only sheet. Net names carry a leading `/` for local labels (`/IN_P`), power nets are bare (`+3V3`, `GND`).

Sheet: root (`bb-amp.kicad_sch`)

Blocks on it: B1 input interface, B2 in-amp, B3 reference/pedestal, B4 output gain stage, B5 output interface, B6 power entry.

Layout of the drawing (left to right, matching the board's own chain, so placement groups at P6 read straight off the schematic): J1 / input pair at the far left; U1 with R1 across its RG pins; the reference cluster (R2, R3, C4, U2A) below U1; U2B with R4/R5 to the right of U1; J2 at the far right; J3 and C2 along the bottom with C1/C3 drawn at their IC supply pins.

Nets that leave a block (these become the placement groups at P6 and the net classes at P5):

net	from	to	note
‘/IN_P‘	J1 pin 1	U1 pin 4 (+IN)	differential pair with ‘/IN_N‘; matched, symmetric, over unbroken B.Cu
‘/IN_N‘	J1 pin 2	U1 pin 1 (-IN)	same
‘GND‘	J1 pin 3, J2 pin 2, J3 pin 2	plane	bias-current return path for both inputs (refdesign D12) - not a convenience pole
‘/VREF_SET‘	R2/R3 junction, C4	U2A + input	high-Z divider node, bypassed by C4
‘/VREF‘	U2A output	U1 pin 6 (REF), R5	must stay below 2 ohm of source impedance (AD8226 Reference Terminal); one node, three loads
‘/AMP1_OUT‘	U1 pin 7 (OUT)	U2B + input	0.21 - 1.05 V over the input range
‘/FB2‘	R4/R5 junction	U2B - input	stage-2 summing node
‘/VOUT‘	U2B output	J2 pin 1	0.113 - 3.037 V into >= 100 k
‘+3V3‘	J3 pin 1	U1 pin 8, U2 pin 8, R2, C1, C2, C3	0.65 mA worst case

Refdes ranges

One sheet, so uniqueness is trivial - the ranges are still fixed here so P4 and any later `schem_refdes` renumber land in the same places, and so a second sheet added later cannot collide.

class	range	assigned on this board
U (ICs)	U1 - U9	U1 AD8226 in-amp; U2 OPA2333 dual (U2A buffer, U2B gain)
R	R1 - R19	R1 RG 1.27 k; R2 121 k, R3 10.0 k divider; R4 24.9 k fb, R5 10.0 k gain return
C	C1 - C19	C1 100 n at U1; C2 10 u bulk; C3 100 n at U2; C4 100 n on ‘/VREF_SET‘
J	J1 - J9	J1 input 3-pole; J2 output 2-pole; J3 power 2-pole
H (mounting)	H1 - H4	none required; optional M3 clearance holes only if the earned outline leaves corner room
‘#PWR‘	‘pwr_base = 100‘ -> ‘#PWR0101...‘	power/GND symbols on the root sheet
‘#FLG‘	‘#FLG0101...‘	one PWR_FLAG on ‘+3V3‘ and one on ‘GND‘ (external rail, no on-board source - ERC needs them)

Values and tolerances the schematic must carry

Calibration removes initial error but not tempco, so tolerance and TCR are split deliberately (`blocks.md` Ruling 3 quantifies each term):

ref	value	tolerance / TCR	why
R1 (RG)	1.27 k	0.1 %, 25 ppm/degC	sets G1 = 39.90; its TCR adds directly to the AD8226's -100 ppm/degC gain drift
R2 / R3	121 k / 10.0 k	0.1 %, 25 ppm/degC	sets the 0.252 V pedestal; ratio TCR mismatch is a 1.1 uV RTI drift term
R4 / R5	24.9 k / 10.0 k	0.1 %, 25 ppm/degC	sets G2 = 3.49; ratio TCR mismatch adds to gain drift
C1, C3, C4	100 nF	X7R, 16 V+	supply and reference bypass
C2	10 uF	X5R/X7R, 16 V+	bulk at the power entry (AD8226 Figure 61)

Absolute tolerance is not what buys accuracy here (zero and span are calibrated downstream per Q7) - TCR is. If P3 finds 1 % / 25 ppm parts cheaper and in stock, they are acceptable everywhere; 100 ppm/degC thick film is NOT, because it would triple the pedestal drift term and add ~50 ppm/degC to the gain drift.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 4 attempt(s); last 2026-08-17T22:28:29, failing 0 of 0.

Review waivers

bb-amp - P4 waivers

The `erc` gate itself is 0 errors / 0 warnings, so nothing is waived there. This file records the disposition of the fresh-context `schematic-reviewer`'s findings (`reports/review-schematic.md`, 0 errors / 5 warnings).

Fixed, not waived (4 of 5)

```
| # | finding | disposition |
|---|---|---|
| W1 | The recorded error budget counted the R2/R3 ratio TCR (1.1 uV) but
omitted the rail term that ratio multiplies.  | FIXED in documentation.  The
pedestal is 0.2519/3.3 = 7.63 % of the rail and reaches the output at gain 1,
so dVout/dVs = 0.0763 V/V (bench B6 measures exactly that) = 548 uV RTI per
volt.  The on-sheet accuracy note now states it and names it the DOMINANT
post-calibration term, ahead of the 1.1 uV the table did count.  |
| W2 | The sheet claimed U2A buffers the pedestal to "< 2 ohm at U1 REF",
unqualified, which this board's own B9 bench contradicts.  | FIXED. The sheet
now carries the measured four-point Zout table (0.43 mohm DC / 0.17 ohm at 60
Hz / 2.86 ohm at 1 kHz / 116 ohm at 41 kHz), says the rule is EXCEEDED above
~700 Hz (350 Hz on the pessimistic ro=2k bracket), and states both why it does
not bite here and what would invalidate that.  |
| W3 | The equations box attributed "Eq.2 allows 1.33 V, G <= 66" to Vcm = 1.65
V; Eq.2 at 1.65 V actually gives 1.49 V and G <= 75.  | FIXED. Two labelled
rows now, with the 1.33 V / G <= 66 pair correctly attributed to the Vcm =
1.7325 V worst corner (rail -5 % AND excitation +5 %).  |
| W4 | `KF128-5.08-2P` carried reference prefix "U" while its 3P sibling carried
"J" - a re-annotate-with-reset would have renamed J2/J3 and broken
constraints.json, parts.json and the P9 CPL. | FIXED in `lib/aiee.kicad_sym`.
The librarian audited all 11 symbols; this was the only instance.  |
```

Waived (1 of 5)

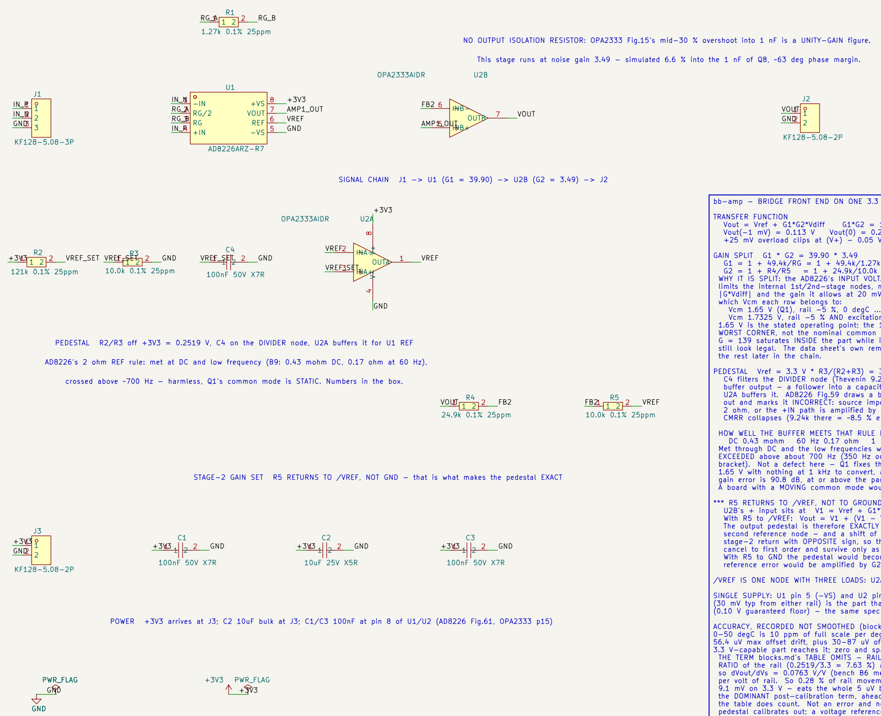
W5 - the connector checklist's "if destructive, flag" item is unmet for J3.

Waived. J3 is the +3V3 power entry on a screw terminal, and the concern is a user wiring it backwards or to the wrong supply. The mitigations the checklist has in mind - reverse polarity protection, a keyed or polarized connector - are PROTECTION, which the `block-only` scope tier excludes outright, so they are not available to this board and their absence is not a finding here. What the board does carry is a silkscreen legend naming the polarity at the terminal. Recorded rather than silently dropped: on a `product`-tier version of this block, reverse-polarity protection at J3 would be REQUIRED, not optional.

Standing limitations noted by the reviewer (not findings)

- AD8226 Table 8's constants are interpolated linearly to 0 C / 50 C from the datasheet's -40 / +25 / +85 rows; the vendor publishes no bench-range values, so the diamond margins (>= 264 mV at full scale) inherit that interpolation.
- OPA2333's DC and low-frequency open-loop output impedance is not published (only 2 kohm at 350 kHz), so bench B9's Zout figures rest on a calibrated model bracket rather than a specification. Already flagged in the part extraction's own OPEN.
- Silkscreen legends, C1/C3 short-loop placement, and the unbroken B.Cu reference under the input pair are P6/P7 items and were not assessable at P4.

The full schematic PDF follows.



NO OUTPUT ISOLATION RESISTOR: OPA2333 Fig.15's mid-30 % overshoot into 1 nF is a UNITY-GAIN figure.

This stage runs at noise gain 3.49 - simulated 6.6 % into the 1 nF of Q8, ~63 deg phase margin.

SIGNAL CHAIN J1 -> U1 (G1 = 39.90) -> U2B (G2 = 3.49) -> J2

PEDESTAL R2/R3 off +3V3 = 0.2519 V, C4 on the DIVIDER node, U2A buffers it for U1 REF

AD8226's 2 ohm REF rule: met at DC and low frequency (B9: 0.43 mohm DC, 0.17 ohm at 60 Hz),

crossed above ~700 Hz - harmless, Q1's common mode is STATIC. Numbers in the box.

STAGE-2 GAIN SET R5 RETURNS TO /VREF, NOT GND - that is what makes the pedestal EXACT

POWER +3V3 arrives at J3; C2 10uF bulk at J3; C1/C3 100nF at pin 8 of U1/U2 (AD8226 Fig.61, OPA2333 p15)

```
bb-amp - BRIDGE FRONT END ON ONE 3.3 V RAIL  (architecture/blocks.md)

TRANSFER FUNCTION
  Vout = Vref + G1*G2*Vdiff     G1*G2 = 139.2 V/V,  Vref = 0.252 V
  Vout(-1 mV) = 0.113 V    Vout(0) = 0.252 V    Vout(+20 mV) = 3.037 V
  +25 mV overload clips at (V+) - 0.05 V and recovers, no fold-back.

GAIN SPLIT  G1 * G2 = 39.90 * 3.49
  G1 = 1 + 49.4k/RG = 1 + 49.4k/1.27k = 39.90  U1 AD8226, R1 across pins 2-3
  G2 = 1 + R4/R5   = 1 + 24.9k/10.0k = 3.49   U2B, OPA2333 half B
 WHY IT IS SPLIT: the AD8226's INPUT VOLTAGE RANGE section (Eq.1-3, Table 8)
 limits the internal 1st/2nd-stage nodes, not the pins.  Eq.2 ceiling on
 |G*Vdiff| and the gain it allows at 20 mV FS (blocks.md Ruling 2) - note
 which Vcm each row belongs to:
   Vcm 1.65 V (Q1), rail -5 %, 0 degC ............. 1.49 V  G <= 75
   Vcm 1.7325 V, rail -5 % AND excitation +5 % ..... 1.33 V  G <= 66
 1.65 V is the stated operating point; the 1.33 V / G <= 66 pair is the
 WORST CORNER, not the nominal common mode.  Either way a single stage at
 G = 139 saturates INSIDE the part while its input and its output both
 still look legal.  The data sheet's own remedy: less gain in the in-amp,
 the rest later in the chain.

PEDESTAL  Vref = 3.3 V * R3/(R2+R3) = 3.3 * 10.0k/131.0k = 0.2519 V
  C4 filters the DIVIDER node (Thevenin 9.24k -> 172 Hz corner), never the
  buffer output - a follower into a capacitive load peaks.
  U2A buffers it.  AD8226 Fig.59 draws a bare divider into REF, crosses it
  out and marks it INCORRECT: source impedance at REF must stay below
  2 ohm, or the +IN path is amplified by 2(50k + Rref)/(100k + Rref) and
  CMRR collapses (9.24k there = ~8.5 % error = ~21 dB of CMRR left).

 HOW WELL THE BUFFER MEETS THAT RULE IS MEASURED, NOT FLAT (bench B9):
   DC 0.43 mohm   60 Hz 0.17 ohm   1 kHz 2.86 ohm   41 kHz 116 ohm
 Met through DC and the low frequencies where the AD8226 specifies CMRR;
 EXCEEDED above about 700 Hz (350 Hz on the pessimistic ro = 2 kohm
 bracket).  Not a defect here - Q1 fixes the common mode at a STATIC
 1.65 V with nothing at 1 kHz to convert, and the residual common-mode
 gain error is 90.8 dB, at or above the part's own 90 dB min at 5 kHz.
 A board with a MOVING common mode would have to rederive this.

*** R5 RETURNS TO /VREF, NOT TO GROUND ***
  U2B's + input sits at  V1 = Vref + G1*Vdiff.
  With R5 to /VREF:  Vout = V1 + (V1 - Vref)*R4/R5 = Vref + G1*G2*Vdiff.
  The output pedestal is therefore EXACTLY Vref - no matched network, no
  second reference node - and a shift of Vref lands on U1 REF and on the
  stage-2 return with OPPOSITE sign, so the reference's own drift and noise
  cancel to first order and survive only as the gain-1 pedestal term.
  With R5 to GND the pedestal would become G2*Vref and every bit of
  reference error would be amplified by G2.  A different, worse board.

/VREF IS ONE NODE WITH THREE LOADS: U2A OUT (1), U1 REF (6), R5.

SINGLE SUPPLY: U1 pin 5 (-VS) and U2 pin 4 (V-) go to GND.  The RRIO OPA2333
(30 mV typ from either rail) is the part that faces the rails, not the AD8226
(0.10 V guaranteed floor) - the same spec that makes a grounded REF unusable.

ACCURACY, RECORDED NOT SMOOTHED (blocks.md Ruling 3): Q7's 5 uV RTI over
0-50 degC is 10 ppm of full scale per degC and is NOT met - 13.9 uV typ /
56.4 uV max offset drift, plus 30-87 uV of gain drift at full scale.  No
3.3 V-capable part reaches it; zero and span are calibrated downstream.
 THE TERM blocks.md's TABLE OMITS - RAIL SENSITIVITY.  The pedestal is a
 RATIO of the rail (0.2519/3.3 = 7.63 %) and reaches the output at gain 1,
 so dVout/dVs = 0.0763 V/V (bench B6 measures exactly that) = 548 uV RTI
 per volt of rail.  So 0.28 % of rail movement AFTER calibration - about
 9.1 mV on 3.3 V - eats the whole 5 uV budget on its own, which makes it
 the DOMINANT post-calibration term, ahead of the 1.1 uV R2/R3 ratio-TCR
 the table does count.  Not an error and not a scope miss (a static
 pedestal calibrates out; a voltage reference is excluded at block-only) -
 it is a requirement on the +3V3 SOURCE, stated here so it is not lost.

NO OUTPUT ISOLATION RESISTOR, AND THAT IS A RESULT, NOT AN OMISSION.  The
OPA2333's ~mid-30 % overshoot into 1 nF (Fig.15) is a UNITY-GAIN figure; this
stage runs at noise gain 3.49.  Simulated into the 1 nF that Q8 allows, the
as-built stage overshoots 6.6 % with about 63 deg of phase margin, so nothing
needs isolating - and at block-only scope, conditioning the data sheet does
not require is out of scope.
```

ai-ee

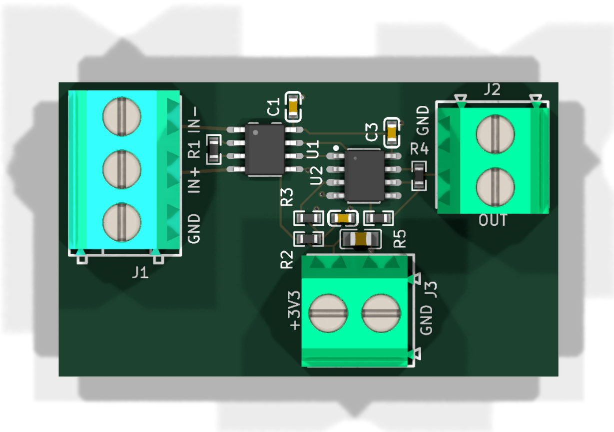
Sheet: /
File: bb-amp.kicad_sch

Title: bb-amp bridge front end AD8226 x OPA2333, G=139.2, 3.3V

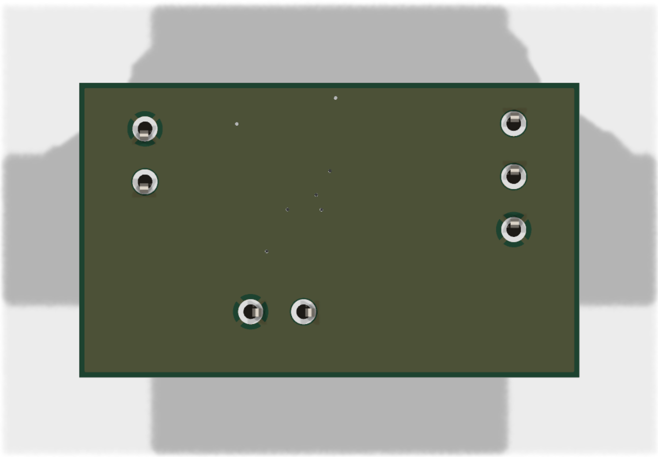
Size: A3	Date: 2026-08-16	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/1

6 Layout

Gate `place` (phase P6): **pass** after 5 attempt(s); last 2026-08-17T22:58:10, failing 0 of 0.
Gate `drc_routed` (phase P7): **pass** after 2 attempt(s); last 2026-08-17T22:57:42, failing 0 of 0.



`reports/renders/bb-amp_top.png`



`reports/renders/bb-amp_bottom.png`

7 Verification

Gate `verify` (phase P8): **pass** after 5 attempt(s); last 2026-08-17T22:57:39, failing 0 of 1.

Verification checks

Check	Status	Findings	Note
`check_return_path`	skipped	0	skipped - no constraints
`check_current`	skipped	0	skipped - no constraints
`check_decoupling`	skipped	0	skipped - no decoupling
`check_diffpair`	pass	0	
`check_creepage`	skipped	0	skipped - no constraints
`check_thermal`	skipped	0	skipped - no constraints
`check_silk`	violations	1	
`check_pdn`	skipped	0	skipped - no constraints/decoupling

Design review of record

bb-amp - P8 board review (adversarial, fresh context)

Scope: mode `learning block-basics: ->` tier `block-only`, binding `canonical`. Nothing below reports an absent feature class the tier excludes (protection, filtering, indicators, test points, config, second rail, mechanical/enclosure). Every claim here was measured from `kicad/bb-amp.kicad_pcb`, the footprint files, the symbol library and my own plots - not read off the pipeline's reports. Where a pipeline claim was checked and held, it is recorded as held.

2 errors, 3 warnings, 1 waiver upheld.

Renders I made for this review:

- `reports/renders/bb-amp_F-SilkS-only.svg` - F.SilkS + Edge.Cuts, nothing else. This is the picture behind finding 1.
- `reports/renders/bb-amp_top-silk-cu.svg` - F.SilkS + F.Cu + Edge.Cuts.
- `reports/renders/bb-amp_bcu-pour-verify.png` - the B.Cu fill polygon drawn from its own 528 vertices, with the input pair (cyan) and its pads (red) overlaid. This is the picture behind verification 2.

—

ERROR 1 - the board carries no connector pole legend at all, and the schematic says twice that it must

`kicad/bb-amp.kicad_pcb` contains exactly 14 pieces of silkscreen text: the 14 refdes. There is no `gr_text` anywhere on the board and no `fp_text` on F.SilkS other than each part's own designator. See `reports/renders/bb-amp_F-SilkS-only.svg`: three screw terminals, seven poles, zero pole markings.

The design's own artifacts demand otherwise. The schematic sheet carries, in its own text, both of these:

- `J1 POLE ORDER: IN- / IN+ / GND - the silk legend must read that way`
- `... THE SILK MUST SAY IN- / IN+ / GND.`

and `requirements.md` s2 promises terminals "silkscreened IN+ / IN- / GND / OUT / +3V3". None of it reached the board.

The only pole-1 indication on any of the three connectors is a 0.21 mm dot (`fp_circle` at footprint-local (-7.62, 5.30) for J1, (-5.08, 5.30) for J2/J3). Placed, J1's dot lands at board (14.000, 24.920) - exactly the corner of the connector's own body outline, i.e. **under the plastic**. After assembly there is no way to tell pole 1 from pole 3 on any terminal.

All three terminals are hand-soldered and hand-wired by the owner (`requirements.md` s7), so the silk is the only thing standing between the user and a miswire. Consequences, worst first:

- **J3 reversed** (pole 1 `+3V3` at x=39.46, pole 2 `GND` at x=44.54, and nothing on the board says so): -3.3 V lands on U1 pin 8 and U2 pin 8. Both parts are past their absolute-maximum supply pin rating and are destroyed. I am *not* reporting the absence of reverse-polarity protection - that is correctly excluded at `block-only`. I am reporting that excluding it makes the legend load-bearing, and the legend is missing.
- **J1 poles 1/2 reversed**: Vdiff is negated, so U1's output wants 0.252 - 39.9*Vdiff. Every positive input drives stage 1 below its guaranteed 0.10 V floor, and the board output sits pinned near the bottom rail across the entire 0-20 mV span. It reads as a dead amplifier, not as a wiring error

  – which is the same presentation as the bias-return failure the workspace has already recorded as symptomless.

- **J1 poles 2/3 reversed**: the sensor's ground lands on IN+; same saturation.
- **J2 reversed**: reading is zero, or the meter's ground is lifted onto /VOUT.

Is there room? Tight, but yes for the critical one. Measured free F.SilkS bands: J1 has a clear 0.95 mm strip outboard at x 13.00..13.95 (rotated text) and a clear ˜2.1 mm strip inboard at x 24.80..26.90 between the body and R1. J2 has only ˜0.65 mm (x 48.6..49.25 inboard, R4 blocks the rest) plus the top edge band y 23.55..24.20 once J2's own refdes is moved. J3 has a ˜0.6 mm band at y 39.50..40.10. So J1 is fixable in place; J2 and J3 realistically want 1.5-2 mm more board on their edges. Under `canonical` binding that is not a violation of the earned-outline principle, it *is* the principle - the board becomes what the placement needs, and a legend the schematic mandates is part of what it needs.

Domain: silk (plus a small `board_edit --outline` grow if J2/J3 are to be labelled properly).

ERROR 2 - every human-facing document states the OPPOSITE pole order to the board

The board is **pole 1 = IN-, pole 2 = IN+, pole 3 = GND** (J1 pad 1 -> /IN_N -> U1.1 `-IN`; pad 2 -> /IN_P -> U1.4 `+IN`). This was a deliberate P6 swap to uncross the pair, and it is right.

But the documents a human reads at H4 all say the other thing:

```
| artifact | says | correct?  |
|---|---|---|
| ‘kicad/bb-amp.kicad_sch‘ (text notes) | IN- / IN+ / GND | YES |
| ‘log/P6-digest.md‘ | pole 1 = IN-, pole 2 = IN+ | YES |
| ‘architecture/blocks.md‘ B1 | "3-pole ...  IN+ / IN- / GND" | **NO** |
| ‘requirements.md‘ s2 (x2) and 9a Q2 | "3 poles (IN+, IN-, GND)" | **NO** |
| ‘reports/design_doc/bb-amp-design-doc.tex/.pdf‘ (4 places) | IN+ / IN- / GND |
**NO** |
```

The design doc was generated 2026-08-16 21:28, i.e. at H1, and was never regenerated after the 2026-08-17 swap. It is the artifact H4 actually reads.

On its own this is documentation drift. Combined with ERROR 1 it is a wiring error waiting to happen: the board has no legend, so the user consults the design doc, and the design doc is wrong. Either fix alone closes the loop; neither has been done.

Domain: review (fix = update blocks.md B1 + requirements.md s2/9a Q2 to match the schematic, and regenerate the design doc).

—

WARNING 3 - verify_all ran 2 of its 8 checks; the verify gate reports 8 of 8

`reports/verify_all.json` (and the identical `kicad/reports/checks/summary.json`) records `coverage.ran = [check_diffpair, check_silk]` and ‘skipped_error = {check_return_path: ”no constraints”, check_current: ”no constraints”, check_decoupling: ”no decoupling”, check_creepage: ”no constraints”, check_thermal: ”no constraints”, check_pdn: ”no constraints/decoupling”}. `verify_all was invoked without` –constraints‘ and without `--decoupling`, even though `kicad/constraints.json` and `kicad/decoupling.json` both exist - and their digests are recorded in `gate-verify.json`'s own `record_result.inputs`.

`reports/gate-verify.json` then reports `coverage.ran` = all eight, `passed` = all eight, `skipped_error` = {}. A reader of the gate cannot see the hole. Per my own brief, a skipped check is a hole, not a pass - so I closed it myself. Re-running all six against the real constraint files:

check	result
check_return_path	pass, but `checked: []` - needs a `high_speed` list, which this board has none of. Vacuous.
check_current	**real pass** - +3V3 needs 0.005 mm, board has 0.2 mm
check_creepage	pass, `checked: []` - the only `voltage_pair` is 0.025 V. Vacuous, and correct: nothing here is above 30 V
check_thermal	pass, `checked: []` - no thermal list. Vacuous, and correct: 2.3 mW total
check_decoupling	**real pass** - C1 1.50 mm / 2.69 nH, C3 1.53 mm / 2.71 nH, C2 bulk 15.99 mm / 12.97 nH, each with its own GND via within 1.11 mm
check_pdn	**real pass** - +3V3 has 3 caps, 10.2 uF, 1 bulk

So **no defect is hiding behind the hole**, and the board's real return-path question is answered directly in verification 2 below. The finding is that the gate's coverage reporting is wrong and would hide a real hole on the next board.

Domain: review.

WARNING 4 - the ERC gate's evidence is stale

`reports/gate-erc.json` (P4) records `sch: sexpr_no_uuid:f12934c5dab8276...`. The current schematic hashes to `sexpr_no_uuid:54a2c346e578e343a...` (I computed it with the pipeline's own `statelib.hash_artifact`). The J1 pad-net rewire at P5/P6 - which required a P5 re-init because `board_update` refuses pad-net rewires - changed the schematic after ERC passed, and ERC was never re-run. The green ERC gate on record does not describe the shipped schematic.

I re-ran it: **0 violations on 54a2c346**. Nothing is hiding here either. Schematic-to-board net parity *is* covered independently - `drc-routed.json` ran with `parity: true` against the current files and found 0.

Domain: review.

WARNING 5 - six clearances sit on the 5 mil fab floor on a board with room to spare

Measured edge-to-edge on F.Cu, pad rotations corrected:

gap	between
0.1287 mm	R2.2 (/VREF_SET) <-> trk +3V3
0.1287 mm	U1.5 (GND) <-> trk +3V3
0.1287 mm	U1.6 (/VREF) <-> trk +3V3
0.1287 mm	U1.7 (/AMP1_OUT) <-> trk +3V3
0.1287 mm (x2)	U2.6 (/FB2) <-> trk /AMP1_OUT
0.1341 mm	R3.2 (GND) <-> trk /VREF_SET

`bb-amp.kicad_dru`'s `aiee_clearance_floor` is 0.1270 mm and `jlc_capabilities.yaml 2layer_1oz min_clearance_mm` is 0.1270 mm, so KiCad DRC passes and `dfm_check` passes all eight sub-checks. This is legal - it is 1.7 um of margin on the fab floor, chosen by the router on a 47.98 x 28.251 mm board carrying 14 parts and 74 track segments. Nothing forced any of them.

Two of the six are on nodes where a short is **silent rather than obvious**: /FB2 shorted to /AMP1_OUT collapses G2 from 3.49 to 1, so the board still amplifies - at 39.9 V/V instead of 139.2 - and presents as a calibration problem rather than a fault. That is worth knowing at bring-up (see below).

Electrical leakage across 0.1287 mm is **not** a concern here and I checked the worst case explicitly: +3V3 to /VREF_SET, the 9.24 k Thevenin divider node. Even at a badly contaminated 100 Mohm surface resistance the pedestal shifts 282 uV, i.e. 2.0 uV RTI, and it is a static term that calibrates out.

Domain: router.

—

Waiver verdict - check_silk / C4 misattribution: UPHELD, with a caveat on the reasoning

Verified independently. The `C4` refdes sits at (42.300, 33.975). U2's silk outline spans x 40.35..43.82, y 29.96..35.12 and the SOP-8 plastic body spans x 40.35..44.25, y 30.09..34.99 - so the label is inside U2's outline and under U2's body. After assembly it is invisible; before assembly it reads as U2's designator while C4's actual body at (40.30, 36.60) carries no label at all. U2's own designator is 2.6 mm to the left, outside the outline, so it is not displaced. KiCad's DRC reports 0 silk findings; `dfm_check`'s silk group passes; JLC places C4 from the CPL, not the silk. **Cosmetic. The waiver stands.**

One caveat, on the reasoning rather than the verdict. The waiver argues that adding ~1.5 mm of board height "fights the earned-outline principle on a board whose binding makes size an OUTPUT of the placement". `reference/build-modes.md` says the opposite: at `canonical` the board *becomes what the placement needs*, and legible silk is part of what it needs. That matters practically, because ERROR 1 forces a silk pass and probably a small grow anyway - C4 should be re-placed in that same pass rather than carried as a standing waiver.

I checked every other refdes on the board by pad-extent distance and found no second misattribution. `C1` at (33.635, 25.900) is the nearest miss - it sits 1.6 mm above U1's silk box - but it is 1.875 mm from C1's own body against 4.18 mm to U1's centre, and it is clear of both plastic bodies after assembly.

—

What I tried to break and could not

These are the load-bearing claims. I measured each from the board rather than believing the reports, and each one held.

1. **The input pair.** True, exactly as claimed.

- `/IN_N`: one F.Cu segment, (19.300, 27.460) -> (30.090, 28.100), width 0.309, 0 vias. Length = sqrt(10.79^2 + 0.64^2) = **10.808963 mm**.
- `/IN_P`: one F.Cu segment, (19.300, 32.540) -> (30.090, 31.900), width 0.309, 0 vias, **10.808963 mm**. Delta **0.000000 mm**.
- Mirroring `/IN_P` about y = 30.0000 reproduces `/IN_N` endpoint for endpoint (60 - 32.540 = 27.460; 60 - 31.900 = 28.100). Deviation 0.
- All 74 track segments on the board are on F.Cu. **Zero B.Cu tracks.** The nearest via to the corridor is 7.13 mm away.
- Full neighbourhood within 4 mm, recomputed independently: 15 items on each leg, pairwise identical, with exactly two departures. `/RG_A`-track to `/IN_N` is 0.9719 mm where `/RG_B`-track to `/IN_P` is 0.9852 mm - a 13 um difference. And the `/IN_P` leg has J1's GND pole at 3.7255 mm where the `/IN_N` leg has a +3V3 track at 3.9618 mm; GND is an *end* pole of a 3-pole terminal and cannot be mirrored, so that one is geometry, not workmanship. The reports' phrase "mirror-identical item by item" is true of the eight nearest items and very nearly true of all fifteen.
- Worth stating because it is a real strength nobody claimed: the two *closest* neighbours to each input trace are the RG pads (U1.2 at 0.1405 mm, R1.1 at 0.6924 mm on the `/IN_N` side; the mirror on the other). An in-amp holds its RG pins at essentially the input pins' own potential, so the nearest copper to each high-impedance input is at ~0 V relative to it. The neighbours are self-guarding as well as symmetric. There is no leakage driving force within 4 mm of either input.

2. The ground reference. True, and I verified it three ways rather than trusting `plane_repair`.

- The B.Cu zone contains exactly **one `filled_polygon`**, 528 vertices. One island, no split.
- Shoelace area of that polygon: **1236.8224 mm2**. `planes_gen.json` recorded 1236.822 at 04:39:18 on the pre-route board (digest `c466f7d5...`); the final board is `2b314ae7...`. Bit-identical - **routing removed nothing**, which follows from there being zero B.Cu copper other than the pour.
- I sampled the pour every 0.25 mm along the centreline of each input trace. The only void on either leg is the first 1.26 mm out of J1 - J1's own through-hole antipad - and it is identical on both legs.
- I sampled 2400 points across the corridor (x 18.0..31.5, y 25.0..35.0) and mirrored each about y = 30.0: **zero asymmetric samples**. The reference under the pair is continuous and mirror-symmetric.
- See `reports/renders/bb-amp_bcu-pour-verify.png`. It also shows something the reports do not: all three GND through-hole pads (J1.3, J2.2, J3.2) carry 4-spoke thermal reliefs into the 1236 mm2 pour, so the hand-soldered terminals are actually solderable. On a board like this that is a real hazard, and it is handled.

3. Topology and pin maps. No correct-but-wrong connection found.

I read the pin numbers out of `lib/aiee.kicad_sym` rather than assuming them: AD8226ARZ = 1 `-IN`, 2 `RG/2`, 3 `RG`, 4 `+IN`, 5 `-VS`, 6 `REF`, 7 `VOUT`, 8 `+VS`; OPA2333AIDR = 1 `OUTA`, 2 `INA-`, 3 `INA+`, 4 `V-`, 5 `INB+`, 6 `INB-`, 7 `OUTB`, 8 `V+`. Both are the real parts. The board's 11 nets land exactly on the intended topology. R5 does return to `/VREF` and not to GND. C4 is on `/VREF_SET`, the divider node, and not on the buffer output. R1 sits across U1 pins 2-3 with `/RG_A` and `/RG_B` at 2.2597 mm each - equal to seven digits - and mirror-placed about y = 30. G1 = 1 + 49.4k/1.27k = 39.898, G2 = 1 + 24.9/10.0 = 3.49, G = 139.24, Vped = 3.3 * 10.0/131.0 = 0.25191 V. R1's long axis is along y, so both of its terminations sit at the same x - the correct orientation against a left-to-right thermal gradient, though on a 2.3 mW board nothing dissipates enough to make it matter.

4. The recorded accuracy miss - re-derived, arithmetic is right, not re-reported. (0.5 + 2/39.9) * 25 = 13.8 uV typ and (2 + 10/39.9) * 25 = 56.3 uV max offset drift; (100 + 25 + 50) ppm/degC * 25 degC * 20 mV = 87 uV max gain drift at full scale, ~30 uV typ; pedestal ratio-TCR 25-50 ppm/degC * 25 degC * 0.252 V / 139.24 = 1.1-2.3 uV RTI. Every term reproduces. On the rail sensitivity: bench b6 measures dVout/dVs = 0.0771, i.e. 0.0771/139.235 = 554 uV RTI per volt, so 5 uV RTI is spent by 9.0 mV = 0.27 % of 3.3 V. The record's 548 uV/V and 0.28 % use the ideal divider ratio 0.0763 and agree to two digits. The reasoning is sound and the miss is honestly stated.

5. Geometry, connectors, edges. Board is one `gr_rect` Edge.Cuts, (13.000, 23.549) - (60.980, 51.800) = 47.980 x 28.251 mm, earned at P6 under `canonical`; no stated dimension exists to drift from. I derived each connector's wire-entry direction from the footprint's own entry notches (local +y) and its placed rotation: J1 (rot -90) opens toward -x, J2 (rot +90) toward +x, J3 (rot 0) toward +y. **All three face off their own board edge**, screws up, with 1.0 mm of board beyond each body. Closest copper to any board edge is 1.1459 mm. There is no F.Cu pour, so nothing couples into the input pair from a top fill.

—

Bring-up - what to check first, in this order

The workspace already records that a missing input bias-current return has no diagnostic symptom. Given this board's actual failure modes, the ordering that matters is:

1. **Before power, ring out J3.** There is no legend and no reverse protection. J3 pole 1 (`+3V3`) is the pole nearer J1; pole 2 (`GND`) is the far one. Confirm continuity from the intended +3V3 wire to U1 pin 8 / U2 pin 8 with the supply off. Reversing J3 destroys both ICs and nothing on the board prevents it. 2. **Power through a current limit and read Iq.** Bench b7 says 529 uA at

3.465 V. Above ~1 mA means a bridge or a reversed part; near zero means no rail. 3. **Measure /VREF before anything else - it is the most diagnostic node on the board.** Expect 0.2519 V. R5's pad 2 at (42.95, 36.60) is an accessible 0603 land on that net. /VREF proves the divider, the buffer and the rail in one reading, and it is completely independent of the input. Do not use the output to judge health first: with J1 open the output can sit near the pedestal *or* walk to a rail, and neither tells you anything. 4. **Split the chain.** Short the two inputs together and tie them to ~1.65 V. /AMP1_OUT (probe U2 pin 5) must read 0.252 V, and J2 must read 0.252 V. That single test separates stage 1 from stage 2 and would immediately catch the /FB2-to-/AMP1_OUT short that WARNING 5 makes possible - G2 would be 1, so J2 would track /AMP1_OUT instead of standing off it. 5. **Probe U1 pins 1 and 4 directly with the sensor connected.** Both must sit near 1.65 V. Either one at a rail means the bias return is missing - J1 pole 3 not landed, or the sensor's excitation return not actually shared. 6. **Prove the input polarity before trusting any reading.** Apply a small known positive Vdiff. The output must move *up* from 0.252 V. If it slams *down* to ~0.03 V, poles 1 and 2 are swapped - and with no legend on the board and a design doc that states the opposite order, this is the single most likely failure a first build will have. It looks exactly like a dead amplifier. 7. **Check the slope, not just that it moves.** 139.24 V/V; 10 mV in should give 1.645 V out. A working-looking 39.9 V/V is the silent signature of item 4's short. 8. **Two things that will look like faults and are not.** (a) Moving the rail 3.135 -> 3.465 V moves the zero by ~25 mV at the output - that is the designed 0.0763 V/V pedestal-to-rail ratio, and it is the dominant post-calibration error term. (b) The chain is open to 41 kHz: bench b5 reads 0.72 uVrms RTI to 1 kHz but 4.05 uVrms to 41 kHz. A wideband meter will show ~5.6x the design noise. Band-limit the reader to ~1 kHz. Also, do not bench this board against Q7's 5 uV RTI - it is recorded as 13.9 uV typ / 56.4 uV worst case and is performing to spec when it misses.

Open

- The AD8226's "REF source impedance below 2 ohm" rule is met only to ~700 Hz. Bench b9 measures the U2A buffer at 0.43 mohm DC, 0.17 ohm at 60 Hz, 2.86 ohm at 1 kHz and 116 ohm at 41 kHz, so above ~700 Hz the buffer's rising Zout - not the AD8226 - sets this board's CMRR (90.8 dB at 1 kHz, ~59 dB at 41 kHz). This is already recorded as a FINDING in b9's bounds and in the schematic notes, and it is defensible *here* only because requirements Q1 pins the common mode statically at 1.65 V with nothing at 1 kHz to reject. I am not re-reporting it. I am flagging it as the one property of this board that is correct because of an assumption in the requirements rather than because of the circuit, and therefore the first thing that breaks if this block is reused with a moving common mode. The schematic already says so; it deserves to survive into the knowledge record.

- What I could not judge from the board and the renders: whether the wire bundles from three terminals on three different edges physically clear each other on a real bench. The geometry looks fine (three separate edges, 1.0 mm of board beyond each body, wire entries all facing outward) but I did not model bend radius or ferrule length.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 1 attempt(s); last 2026-08-17T23:00:32, failing 0 of 0. DFM findings by severity: none.

Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
100nF 50V X7R	C1,C3,C4	C0603	C14663
10uF 25V X5R	C2	C0805	C15850
1.27k 0.1% 25ppm	R1	R0603	C861195
121k 0.1% 25ppm	R2	R0603	C861095
10.0k 0.1% 25ppm	R3,R5	R0603	C95204
24.9k 0.1% 25ppm	R4	R0603	C136967
AD8226ARZ-R7	U1	SOIC-8_L5.0-W4.0-P1.27-LS6.0-BL	C34250
OPA2333AIDR	U2	SOP-8_L4.9-W3.9-P1.27-LS6.0-BL	C38732

2 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake - digest

- Mode ‘learning block-basics:‘ -> scope ‘block-only‘, binding ‘canonical‘
(geometry is an OUTPUT; P5 opens ‘--outline auto‘).
- ‘requirements.md‘ written; ‘check_requirements.py‘ pass, 0 violations,
sections 1-9, mode named.  (opus/xhigh - fable tier unavailable, logged.)
- 10 questions batched in one round, all answered, appended as section 9a.
- Design-changing answers:  excitation 3.3 V shared-ground (CM 1.65 V, so a
single-supply in-amp stays viable); output span ~0.05-3.25 V with a positive
pedestal (the brief’s literal 0-3.3 V loses to its own rail); 12-bit usable
calibrated (~5 uV RTI = drift + noise); flat within 1% at 1 kHz.
- Derived and binding:  ~7 kHz -3 dB corner => ~1 MHz gain-bandwidth at gain
~150.  Dominant part constraint; rules out low-power zero-drift in-amps.
- Carried to P2:  the pedestal needs a low-impedance REF drive, or an in-amp
with a high-impedance REF - a divider on a 3-op-amp REF kills CMRR.
- Owner delegated remaining judgment; H1-H4 are records, run stops at P9.
- Safety:  no flags apply (3.3 V SELV bench board, milliamps).
```

P1

```
# P1 Research - digest

- Roster:  component-scout (‘inamp‘) + reference-design (‘bridge-front-end‘).
Power-architect skipped (trivially powered); interface-spec skipped (screw
terminals are not standards-bound).
- REF-pin impedance is the architecture fork, cross-checked TI+ADI: classic
3-op-amp in-amps (AD8226/INA333/INA828) need REF driven low-Z (AD8226 Fig.59
marks the bare divider "INCORRECT"); ICF parts (AD8237/AD8420) have 100M-800M
REF and tolerate a divider.
- Bandwidth vs drift squeeze at 3.3 V: INA333 makes drift (0.1 uV/C) but only
~2.2 kHz at G~150; INA828 has the bandwidth but needs 4.5 V min.  AD8226
(2.2-36 V, ~13.6 kHz at G=147, CMRR 120 dB min, 22-24 nV/rtHz) clears both.
- Discrete 3-op-amp killed quantitatively:  TI SBOA582 Eq.7 puts 0.01%
resistors
~44 dB short of the ~118 dB this board needs.
- RECONCILED BY ORCHESTRATOR: the scout’s grounded-REF split dies on AD8226’s
```

(V-)+0.1 V output floor - at G1=100 the bottom 2 mV of span sits in stage-1 saturation. A positive buffered REF is unavoidable; handed to P2 with the math.
- P2 must verify: BW-at-gain (interpolated), real RTI drift (VOSI+VOSO/G), and the Vout-vs-Vcm diamond plot at Vs=3.3 V.

P2

P2 Architecture - digest

- J1 3-pole -> U1 AD8226 G1=39.9 (RG 1.27k, REF on a buffered 0.252 V node) -> U2B OPA2333-class G2=3.49, gain resistor RETURNED TO /VREF -> J2; U2A buffers the divider. Vout = 0.252 + 139.2*Vdiff. 14 parts, 2 ICs, one sheet.
- f-3dB 41 kHz (5.9x spec), 0.65 mA, 2.3 mW, JLC2313_1.6 2L, ~USD 6.6/board.
- The split is forced by the DIAMOND PLOT and the claim was VERIFIED on the page: printed p.19 gives Vout = G*Vdiff + Vref, so Eq.2 becomes Vcm + |Vout-Vref|/2 < Vs - V+LIMIT with NO gain term - Fig.9 (G=1) and Fig.12 (G=100) are the same hexagon. Max single-stage gain 66-95, not 147.
- ACCURACY MISSED AND RECORDED: 13.9 uV typ / 56.4 uV max offset drift over +-25 C vs a 5 uV budget, plus 30-87 uV gain drift at FS. That budget is 10 ppm/degC of full scale; no 3.3 V-capable part reaches it. INA333 alternative costed and rejected on evidence. Mode relaxed nothing.
- Research: 4 tasks, 2 rounds, 6/6 budget. Round 1 seeded a domain the library
did not hold, but was witnessed by the wrong part - 7 of 13 refuted. Round 2 re-witnessed from the AD8226 via an allowlisted Farnell mirror - 9 of 9 verified. Net 16 verified records + 2 new checklists (‘inamp‘, ‘in‘).
- Coverage exit 0: 2 slots, 0 gaps, 11 classes provisional (the correct ceiling - owner promotion makes approved, bring-up makes proven).
- 8 sim benches specified with numeric windows for the P8 ‘sim‘ gate.

P3

P3 Parts + Library - digest

- BOM 15 refdes / 11 LCSC lines, USD 5.70/board at qty 5. All five gain and pedestal resistors 0.1% 25 ppm/degC thin film - exactly what the P2 error budget assumed, no downgrade.
- R6 ADDED, overruling blocks.md B5: OPA2333 Fig.15 shows ~mid-30% overshoot into 1 nF and Q8 allows 1 nF of output cable. 100R provisional; the B4 bench settles it at P8.
- Connectors KF128-5.08, hand-soldered THT. J1 is a single 3-pole body so IN+/IN- junctions are identically built - the dominant 5 uV error term. The research's own witness was the CABLE half of a pluggable pair; caught here.
- Extractions validate (8 pins, land patterns, 10 layout_notes each). The AD8226 extraction independently cross-checked every number blocks.md cited - zero contradictions.
- Library: 11 symbols, 7 footprints, all catalog pulls, 33 pins retyped. fp_verify 4/4 PASS after one fix - J2/J3 pulled a 1.6 mm drill where the datasheet and J1's own entry both say 1.4 mm. Corrected.
- Coverage exit 0: 4 slots, 0 gaps; part slots covered, block slots provisional. 2 of 6 research budget unspent.
- Gaps recorded not smoothed: U1 pad size unverifiable (no vendor land pattern exists); B1's INL sim window is not a datasheet number.

P4

P4 Schematic + sim - digest

- ONE sheet (recorded): 14 refdes, 11 nets, 41/41 pins. ERC 0/0, netlist_audit
0 violations, regenerate-and-compare identical. Gate ‘erc‘ PASS, committed.
- Load-bearing net verified in the NETLIST by a fresh reviewer, not the drawing:
R5 returns to /VREF, U1 pin 6 REF driven by U2A as a true unity buffer, both OPA2333 units placed with power pins present.
- R6 REMOVED, reversing my own P3 call. Fig.15’s ~35% overshoot is a UNITY-GAIN
figure; U2B runs at noise gain 3.49. The bench reproducing Fig.15 exactly (32% at 1 nF unity) gives 6.6% for the as-built stage with none fitted. Board measures better without it: slope 139.235, -5% rail FS 3.0221 -> 3.02513 V, span 99.974%.
- 11 benches / 76 bounds green. Six seeded defects verified to TRIP the gate; the analyst caught two defects in its own models; the removal exposed two stale bench constants - the sidecars caught their own author.
- Reviewer 0 errors / 5 warnings, nothing electrically wrong. Four FIXED (rail term missing from the error budget, an unqualified <2 ohm REF claim its own bench contradicts, a misattributed diamond number, a connector prefix that would break on re-annotate); one waived (J3 polarity - protection excluded).
- Text edits proven non-electrical: canonical netlist digest byte-identical, plus --compare showing 0 diffs and 0 renames.

P5

P5 Board setup - digest

- Ran INLINE. Sidecars placed beside the board (constraints, decoupling, parts).
- ‘board_init --outline auto --margin 2 --mounting-holes 0‘: parity 0, setup_violations 0, clean true. 14 components, 11 nets, stackup JLC2313_1.6 2L. The 30 unconnected are P7’s.
- Fab floors REPORTED not assumed (2layer_1oz, all ERROR): clearance 0.127, track 0.127, via 0.6, hole 0.3, annular 0.15, hole-hole 0.5, edge 0.3 mm.
- Provisional outline 41.0 x 57.7 mm - generous and NOT the board. Geometry is an OUTPUT: P6 places, then ‘board_edit --outline fit‘ earns the size.
- DEFECT CAUGHT BEFORE IT SHIPPED: rules_gen defaults a diff_pairs entry with no ‘impedance_ohm‘ to 90 ohm and solves the netclass WIDTH from it - this DC-1 kHz 20 mV sensor pair would have routed at 1.3743 mm on a board with no controlled impedance anywhere, and the router traces at class width. Fixed by declaring the impedance the pair ACTUALLY HAS at the width we want (150 ohm -> 0.3089 mm), with an in-file note that it is a consequence, not a target. Deleting the entry was rejected: knowledge retrieval keys the 8 verified interface records off diff_pairs[].base (confirmed - P6 select pulled 16).
- ‘rules_gen‘: 11 rules, netclass Diff150, +3V3 stays Default 0.2 mm.

P6

P6 Placement - digest

- Gate ‘place‘ PASS, 0 failing across all 5 legality legs. DRC --severity-all + parity: 30 findings, ALL unconnected (unrouted) - 0 copper, 0 silk, 0 courtyard, 0 parity. check_decoupling pass. Route probe on a scratch copy: completion 1.00, 0 unrouted of 30.
- BOARD SIZE EARNED: 47.98 x 28.251 mm (content 45.980 x 26.250, margin 1.0). Geometry was an OUTPUT throughout; no dimension was ever a target.
- My P5 digest misread ‘outline_bbox‘ as a size. It is min/max CORNERS: the provisional room was 28.02 x 44.69, not 41.0 x 57.7. The canonical layout needs ~46 mm of width, so the room had to be grown. Order is

PLACE -> GROW -> FIT, because fit CLIPS and growing a fresh shelf pack refuses.
- The annealer's 4 candidates were all REJECTED: movable_clusters 2 of 5, so every candidate HPWL was dominated by connectors pinned to a provisional outline. Canonical tile hand-built instead, 2 of 8 edit iterations used.
- J1 PIN SWAP (pole 1 = IN-, pole 2 = IN+) taken on the agent's finding that the
connector's pole order and the AD8226's pin order are reversed, forcing one crossing under all 8 rotation/edge combinations. Cost a P5 re-init because board_update refuses pad-net rewires; ops trail made the replay exact.
Result: pair HPWL 15.23 -> 11.43 mm/leg, crossings 6 -> 5, length delta 0.000000 mm, mirror axis y=30.0000 exact. Verified by machine: two straight segments, ZERO vias, 0 non-unconnected DRC findings, unconnected 30 -> 28.
- Decoupling: C1 1.50 mm, C3 1.53 mm (both straight-shot), C2 bulk 15.99 mm under the 20 mm warn - and vendor-correct, the 10 uF is explicitly allowed farther out. /VREF is a 4.11 mm straight hop. J1-to-J2 ~35 mm, pour between.

P7

P7 Routing - digest

- Gate 'drc_routed' PASS, 0 failing. DRC 0 errors / 0 warnings at any severity,
0 unconnected, all 30 connections closed. Completion 1.00. Committed.
- Freerouting rung 1 only (mp:20, 2 passes, 0 unrouted of 23), **0 of the 2-retry budget used**, no KRT fallback, no placement_adjust_request.
- INPUT PAIR HELD, measured on the final board: /IN_N and /IN_P both 10.808963 mm, delta 0.000000 mm, width 0.3089 mm both, **0 vias**, one F.Cu segment each, mirror deviation about y=30.0000 of 0.000000 mm. Neighbourhood mirror-identical item by item. Zero vias in the input corridor (nearest 7.13 mm). check_diffpair PASS, skew 0.0.
- Router OVERRULED 'route_critical --only diff': that path requires a COUPLED pair at 0.3 mm gap and would have converged both legs onto y=30.0 into R1's pads. Laid two straight segments instead; Freerouting preserved them byte-exact. Third sighting of the diff_pairs symmetry-vs-impedance conflation.
- Reference is intact, measured not inferred: bottom pour is ONE component, 1236.8224 mm2, split false, **bit-identical to what planes_gen created before any copper was laid**. Zero B.Cu tracks anywhere - every signal on F.Cu.
7 vias total, all GND pad-stitches; stitch_vias added 0 (57 of 65 area candidates rejected single_layer_contact - a stitch via on a one-sided 2L pour is an antenna).
- 'route_cleanup': dry-run 0 ops, live 0 ops, drc_before == drc_after. The V13 dangling-pass defect did NOT recur on its exact signature (7 GND stubs whose connectivity runs through the pour).
- Aggressor separation held: /VOUT to /IN_N 13.51 mm, to /IN_P 12.98 mm; J1 to J2 terminals 35.82 mm - no routing detour undid P6.
- Two errors of mine in the brief, both caught: I cited the REFUTED in-aggressor-separation record as governing, and I named /AMP2_OUT, a net deleted with R6 at P4.

P8

P8 Verification - digest

- All five gates green: erc PASS, place PASS, drc_routed PASS (DRC 0/0, completion 1.00), verify PASS (8/8 checks ran, 8/8 passed), sim PASS (11 benches, 76 bounds). Advisory legs check_irdrop and check_pdn_z: pass.
- check_diffpair initially FAILED the gate demanding sub-0.75 mm coupling over a deliberately mirror-symmetric pair. Fixed the PREMISE, not the symptom: max_uncoupled_mm 5.0 -> 12.0 with a note that this pair has no coupling

```
requirement.  FOURTH sighting of the diff_pairs symmetry-vs-impedance
conflation on this board (P2 slot typing, P5 netclass width, P7 route_critical,
P8 check_diffpair).
- Fresh verify-reviewer found 2 ERRORS, both fixed rather than waived:
1.  THE BOARD HAD NO CONNECTOR POLE LEGEND AT ALL - 14 silk items, all refdes,
7 hand-wired poles unmarked.  J3 reversed destroys both ICs; J1 reversed
reads as a dead amplifier.  Invisible to every gate.  7 legends now placed
outside the housings, board NOT grown, DRC 0.
2.  Every human-facing doc stated the OPPOSITE pole order (requirements.md
s2 + 9a Q2, blocks.md B1 + diagram, the H1 design-doc PDF). The P6 swap
reached the netlist and schematic but not upstream.  All corrected.
- Reviewer independently VERIFIED the board's two central claims by
measurement,
not by trusting reports:  input pair one F.Cu segment per leg, 0 vias,
10.808963 mm each, delta 0.000000, mirroring /IN_P about y=30.0000 reproduces
/IN_N endpoint for endpoint; B.Cu pour one polygon, 528 vertices,
1236.8224 mm2, bit-identical to its pre-routing area, 2400 mirrored corridor
samples with zero asymmetry.
- 1 warning waived with reasoning:  C4's refdes reads as U2's label; silk_place
finds no collision-free position at 0.5 mm and KiCad's own DRC reports 0 silk
findings.
- Tooling defects logged:  verify_all without sidecars runs 2 of 8 checks while
the gate reports 8/8 coverage; render.py's out-dir does not match
report_gen's search ladder, silently dropping the layout section.
```

P9

```
# P9 DFM + run close - digest

- Gate `dfm` PASS, 0 violations at any severity.  Re-run standalone with the
polarity oracle (schematic) and the BOM leg (parts.json) both supplied so it
did not pass vacuously:  still 0.
- Fab package written to fab/:  gerbers + drill (zipped), bb-amp-pos.csv,
BOM.csv, BOM-full.csv, CPL.csv.
- Assembly split checked, not assumed:  11 SMT parts in CPL.csv; J1/J2/J3
absent
from both CPL and the assembler BOM because they are `hand_install`, and
present only in BOM-full.csv with per-part instructions.  U1/U2 rotations came
through as 180 deg against 270 in the raw position file - the jlc_rotations
correction, which is the catcher for a part mounted backwards.
- BOM-full carries the notes that stop a future reader "fixing" the design:
C4 marked "NOT a supply-pin cap - do not relocate to U2A's output", and J1
marked that the bias-current return runs through the 350 ohm sensor source
itself so no bleeder resistors belong there.
- LEARNINGS: 17 entries compiled to learnings/queue.yaml, all `pending` - 14
from the agents, 3 orchestrator-level (the diff_pairs conflation across four
phases; a late pin change must sweep every artifact that states pin order;
KiCad draws text-box overflow instead of clipping it).  0 orphans,
0 malformed.  Promotion is a separate owner pass.
- RUN STOPS HERE. block-only default:  ordering is a separate owner decision.
No H5, no release attestation, no money.  Everything needed to order is in
boards/bb-amp/fab/.
```

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-16T22:46:35	erc	pass	0/0
2026-08-16T23:11:38	erc	pass	0/0

Timestamp	Gate	Status	Failing/Total
2026-08-16T23:41:23	erc	pass	0/0
2026-08-17T21:06:36	place	pass	0/0
2026-08-17T21:11:15	place	pass	0/0
2026-08-17T21:26:32	place	pass	0/0
2026-08-17T21:32:00	place	pass	0/0
2026-08-17T21:52:01	drc_routed	pass	0/0
2026-08-17T21:52:51	verify	fail	1/2
2026-08-17T21:54:55	verify	pass	0/1
2026-08-17T21:57:29	verify	pass	0/1
2026-08-17T21:57:47	verify	pass	0/1
2026-08-17T21:58:31	sim	pass	0/0
2026-08-17T22:28:29	erc	pass	0/0
2026-08-17T22:57:39	verify	pass	0/1
2026-08-17T22:57:42	drc_routed	pass	0/0
2026-08-17T22:58:10	place	pass	0/0
2026-08-17T23:00:32	dfm	pass	0/0

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	skipped	2026-08-16T21:29:20	H1 presented as a record rather than a blocking hold, per the owner's standing instruction of 2026-08-16 ('Make decisions yourself based on what will give the best learning for the basic system') while three boards run in parallel. Recorded as SKIPPED and deliberately not as APPROVED: no human has ruled on this content, and the status vocabulary has no 'presented under delegation' value. The run proceeds to P3. Nothing before P10 is irreversible and the block-only default stops the run at P9, so no money-spending step is reached without an explicit owner decision. Design doc: reports/design_doc/bb-amp-design-doc.pdf
2	P4	skipped	2026-08-16T23:42:10	H2 presented as a record rather than a blocking hold, per the owner standing instruction of 2026-08-16. Recorded as SKIPPED and deliberately not APPROVED: no human has ruled on this content. Artifacts: reports/schematic.pdf, reports/review-schematic.md (0 errors / 5 warnings), reports/erc-waivers.md (4 fixed, 1 waived), reports/sim.json (11 benches green). Gate erc PASS and committed.
3	P6	not reached		

Id	Phase	Status	When	Note
4	P8	skipped	2026-08-17T22:59:29	H4 presented as a record per the owner standing delegation of 2026-08-16; NOT an owner approval. All five gates green. Reviewer found 2 errors (no connector pole legend; every human-facing doc stated the opposite pole order) - both FIXED, not waived. 1 warning waived (C4 refdes reads as U2's, no collision-free position exists). Artifacts: reports/review-board.md, verify-waivers.json, renders/, design_doc/.
5	P10	not reached		

10 Artifact Index

File	Size	Note
`state.json`	126,735 B	
`kicad/bb-amp.kicad_pcb`	104,993 B	
`kicad/bb-amp.kicad_sch`	81,987 B	
`fab/bb-amp_gerbers.zip`	16,312 B	
`fab/BOM.csv`	357 B	
`fab/CPL.csv`	403 B	
`brief/brief.md`	575 B	
`requirements.md`	18,496 B	
`architecture/blocks.md`	27,378 B	
`architecture/constraints.json`	5,190 B	
`architecture/power_tree.md`	4,310 B	
`architecture/sheets.md`	4,394 B	
`architecture/stackup.md`	3,813 B	
`reports/aggressor-separation.json`	225 B	
`reports/board_edit-fit.json`	1,780 B	
`reports/board_edit-open.json`	8,187 B	
`reports/board_init.json`	1,596 B	
`reports/bom_cpl.json`	9,484 B	
`reports/check_decoupling-p6.json`	1,059 B	
`reports/check_diffpair-p7.json`	886 B	
`reports/check_irdrop.json`	2,079 B	
`reports/check_pdn_z.json`	893 B	
`reports/check_silk-p8-fix.json`	1,291 B	
`reports/dfm_check.json`	1,814 B	
`reports/drc-p6.json`	23,132 B	
`reports/drc-p7-critical.json`	17,891 B	
`reports/drc-p7-pair.json`	20,151 B	
`reports/drc-p8-silkfix.json`	588 B	
`reports/drc-routed.json`	585 B	
`reports/erc-waivers.md`	3,194 B	
`reports/fab_export.json`	2,215 B	
`reports/gate-dfm.json`	1,598 B	
`reports/gate-drc_routed.json`	1,943 B	
`reports/gate-erc.json`	871 B	
`reports/gate-place.json`	1,483 B	
`reports/gate-sim.json`	735 B	
`reports/gate-verify.json`	2,201 B	
`reports/input-pair-symmetry.json`	2,188 B	
`reports/place_anneal.json`	4,157 B	

File	Size	Note
reports/place_edit-canonical.json	1,788 B	
reports/place_edit-canonical2.json	1,788 B	
reports/place_edit-silk-legend.json	1,069 B	
reports/place_metrics-pre.json	4,956 B	
reports/place_metrics.json	4,772 B	
reports/place_seed.json	1,081 B	
reports/plane_repair-flag.json	907 B	
reports/planes_gen.json	857 B	
reports/render-final.json	745 B	
reports/render-h4.json	437 B	
reports/render-p6.json	588 B	
reports/render-p6b.json	435 B	
reports/render-p8.json	435 B	
reports/review-board.json	11,415 B	
reports/review-board.md	20,561 B	
reports/review-schematic.json	3,340 B	
reports/review-schematic.md	13,757 B	
reports/route_auto.json	791 B	
reports/route_cleanup-dry.json	750 B	
reports/route_cleanup.json	913 B	
reports/route_edit-critical-stubs.json	515 B	
reports/route_edit-input-pair.json	407 B	
reports/rules_gen.json	1,042 B	
reports/schematic.pdf	90,783 B	
reports/silk_place-c4.json	1,369 B	
reports/silk_place.json	2,108 B	
reports/sim.json	8,948 B	
reports/stitch_vias-dry.json	1,631 B	
reports/stitch_vias.json	1,632 B	
reports/verify-waivers.json	1,678 B	
reports/verify_all-p8-fix.json	3,844 B	
reports/verify_all.json	3,439 B	